

LSC

Alternative Wave: A Reproducible Multi-Messenger Architecture for the Origin of KM3-230213A

Evidence-constrained review manuscript

19 September 2026

Historical LSC 6.5.0 status: NOT_YET_VALIDATED. LSC-AW is a separate post-freeze lineage.

Editorial Status and Scope

This volume is a scientific research manuscript, not a claim of discovery. It distinguishes direct observation, standard interpretation, active published hypotheses, preprint hypotheses, and LSC sandbox proposals. The purpose is to make competing explanations of KM3-230213A comparable under common detector and multi-messenger constraints. Historical LSC gallium work is included only for methodological lineage and version boundaries.

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Abstract

KM3-230213A is the highest-energy cosmic-neutrino event reported to date. KM3NeT reconstructed a through-going muon with an energy near 120_{-60}^{+110} PeV and, under an E_ν^{-2} incident spectrum, inferred a median parent-neutrino energy near 220 PeV with a broad 72 PeV–2.6 EeV 90% interval. This book develops LSC Alternative Wave (LSC-AW) as a reproducible model-comparison architecture rather than a preferred exotic explanation. Conventional diffuse and transient astrophysical sources, cosmogenic UHECR neutrinos, primordial-black-hole evaporation, heavy relic decay, sterile/NSI/dark propagation, and quantum-geometry hypotheses are placed in a common source–propagation–detector framework and assessed against KM3NeT, IceCube, Auger, electromagnetic constraints, rate calculations, causality, and prospective predictions. The strongest conclusion is methodological: present data do not uniquely identify the source, minimal four-dimensional PBH evaporation is under strong multi-messenger pressure, and quantum-geometry ideas remain sandbox hypotheses until they produce normalized spectra and rates. LSC-AW inherits from the historical gallium LSC programme a freeze-and-falsify discipline but does not transfer its fitted parameter into the UHE sector.

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The Neutrino Event in Plain Language

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A neutrino telescope does not photograph a neutrino. It observes secondary charged particles created when a neutrino interacts in, or near, an instrumented volume. In the case of KM3-230213A, the decisive object was a long, exceptionally bright muon track crossing the partially built ARCA detector in the Mediterranean Sea. The event occurred on 13 February 2023 at 01:16:47 UTC. Twenty-one detection lines were operating. The collaboration reported 28,086 photomultiplier hits and 3,672 triggered photomultipliers, roughly 35 percent of the active devices. The reconstructed direction was almost horizontal, about 0.6 degrees above the local horizon. These are detector-level facts; they are more secure than any interpretation of the source.

The collaboration estimated the muon energy at the detector to be about 120 PeV, with asymmetric uncertainty of +110/-60 PeV. That number is already extraordinary: one petaelectronvolt is 10^{15} electronvolts. Yet the neutrino that made the muon had to be more energetic than the muon itself. Converting the track information into a parent-neutrino energy requires simulation, interaction physics, Earth propagation, and an assumed incident spectrum. Under an E_ν^{-2} spectrum, the median parent energy is about 220 PeV, but the 90 percent interval is very broad, from about 72 PeV to 2.6 EeV. The phrase “a 220 PeV neutrino” is therefore convenient shorthand, not a direct calorimetric measurement.

Why does one event matter? Because the known astrophysical neutrino flux falls steeply with energy. An event so far above the previously well-populated range asks whether the same source population simply produced a rare tail, whether a new source population emerges, whether ultra-high-energy cosmic rays produced the neutrino during propagation, or whether something more exotic is involved. One event cannot choose among those possibilities by itself, but it can make previously academic model classes observationally relevant.

The LSC Alternative Wave programme begins from that asymmetry between a spectacular observation and weak source identification. It does not start by declaring a primordial black hole, sterile neutrino, dark particle, or quantum bounce. It starts with a common inference language: what did the detector see; what did we infer; what other detectors did not see; what

electromagnetic and cosmic-ray signals should accompany each mechanism; and what future observation would discriminate among them.

A recurring theme will be restraint. The event is evidence that nature produces neutrinos in the ultra-high-energy regime. It is not evidence, by itself, for any specific source class. The most productive use of the anomaly is therefore comparative: force every model to explain the same event under the same detector response and the same null constraints.

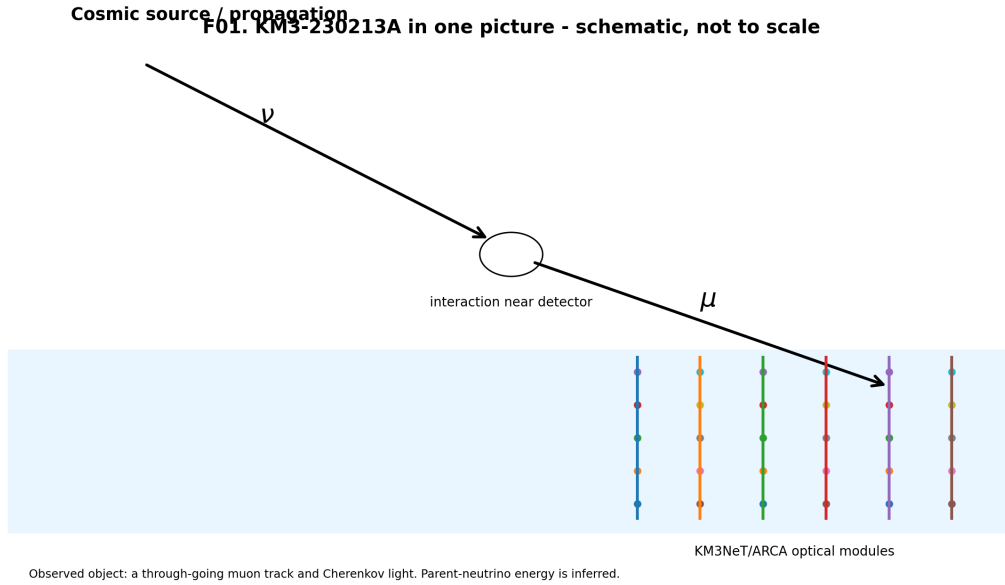


Figure 1.1: detector schematic.

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What KM3NeT Actually Measured

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KM3NeT/ARCA is an underwater Cherenkov array. Charged particles moving faster than the phase velocity of light in seawater emit Cherenkov photons, which are detected by photomultipliers distributed through vertical detection units. For a through-going muon, direction reconstruction relies on the timing and positions of early hits. Energy reconstruction relies on how much light the muon produces through stochastic losses such as bremsstrahlung, pair production, and photonuclear interactions.

For KM3-230213A, the collaboration used the number of triggered photomultipliers as a robust energy estimator because the event was so bright that sensors near the trajectory were saturated. Monte Carlo samples at several muon energies were used to map the observed trigger multiplicity onto a frequentist confidence interval for the true muon energy. Systematic uncertainty is dominated by the optical properties of seawater, especially the absorption length. The published 90 percent muon-energy interval is approximately 35–380 PeV.

Direction reconstruction places the track near the horizon, with J2000 equatorial coordinates around right ascension 94.3 degrees and declination -7.8 degrees. The quoted containment radii are approximately 1.5 degrees at 68 percent and 3.0 degrees at 99 percent, dominated by systematic uncertainty in the absolute orientation of the detector at the time of the event. The collaboration has subsequently worked on improved positioning and orientation calibration. This matters for counterpart searches: a one-degree change in a crowded extragalactic sky can alter which catalogued objects lie inside the preferred region.

Atmospheric explanations are strongly disfavoured. A conventional atmospheric muon arriving from the reconstructed direction would have to traverse an implausibly large water-equivalent column. Atmospheric neutrinos are not impossible in principle, but their expected rate above 100 PeV is tiny. The event is therefore securely classified as cosmic in origin at the level relevant for this book.

The public KM3NeT repository is unusually valuable. It contains event hit information, source data for selected figures, a sky-averaged all-flavour effective area for the bright-track selection, and notebooks reproducing selected published outputs. The official expected-events notebook uses a 335-day livetime for the 19/21-line configurations and integrates a public effective area against assumed flux models. That reproducibility package does not expose every internal calibration or

all detector Monte Carlo, but it is sufficient to anchor several high-level checks and to prevent a model comparison from relying only on digitized journal plots.

A crucial distinction is that the detector measured a track, not a source. The source class, the parent spectrum, and the neutrino production mechanism all remain in the inference layer. LSC-AW therefore stores detector observables separately from source-level variables. This seemingly bureaucratic choice is a scientific safeguard: it prevents an attractive source story from being mistaken for an observed fact.

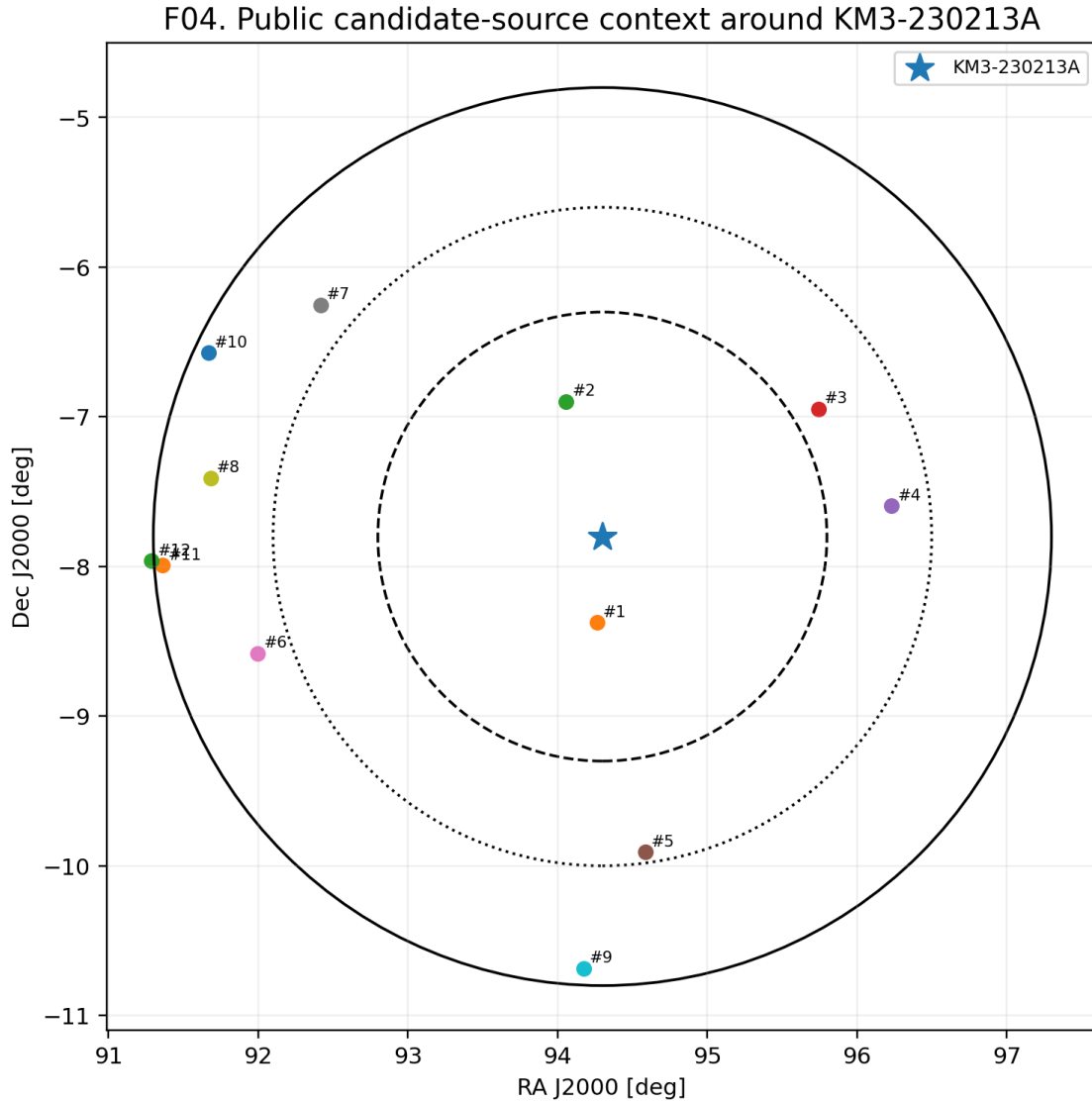


Figure 2.1: F04 skymap.

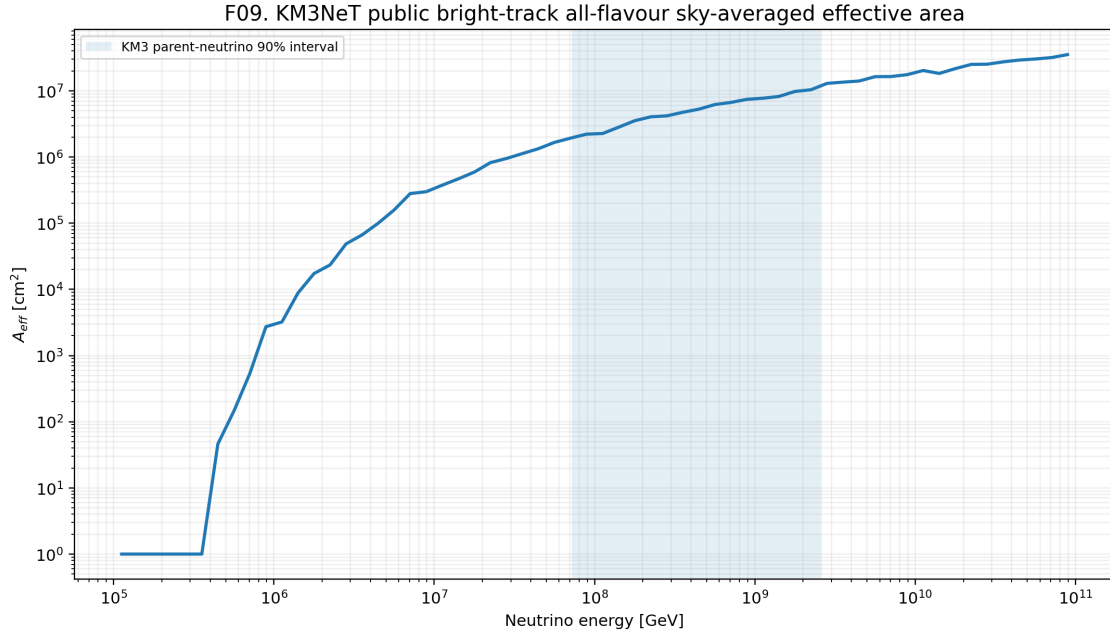


Figure 2.2: F09 effective area.

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Why 220 PeV Is Not an Exact Energy

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The parent-neutrino energy is a posterior-like quantity conditioned on a detector response and an assumed incident spectrum. The published KM3NeT statement is explicit: for an incoming spectrum proportional to E_ν^{-2} , simulated events compatible with the observed muon yield a median parent energy near 220 PeV, a 68 percent interval of roughly 110–790 PeV, and a 90 percent interval of roughly 72 PeV–2.6 EeV.

This dependence can be written schematically as

$$p(E_\nu \mid y, M) \propto p(y \mid E_\nu) \Phi_M(E_\nu),$$

where y denotes detector observables, $p(y \mid E_\nu)$ is the response likelihood, and Φ_M is the spectrum predicted or assumed under model M . A harder spectrum gives relatively more weight to the high-energy tail than a softer spectrum. A narrow line-like model would require its own reweighting rather than inheriting the E^{-2} posterior unchanged.

This point is not cosmetic. Several exotic explanations proposed after the event have sharply structured spectra: heavy dark-matter decay can produce edges and bumps; a terminal primordial-black-hole burst has a rapidly time-evolving spectrum; some sterile or dark-sector models predict detector-dependent conversion rather than a simple power law. Comparing those models to a single fixed value of 220 PeV would mix an inference made under one spectrum with predictions from a different spectrum.

In the LSC-AW architecture, the energy layer therefore has three objects: the measured muon-energy information; the model-conditioned neutrino-energy distribution; and the source-level spectrum. Plots must show the full relevant interval, not just a vertical line at 220 PeV. Likelihood evaluations should fold the predicted spectrum through the detector response whenever that response is available.

The same discipline applies to language. It is acceptable to say that KM3-230213A is commonly described as a roughly 220 PeV neutrino event. It is more precise to say that KM3NeT observed a 120 PeV-scale muon and inferred a median parent-neutrino energy near 220 PeV under an E^{-2} incident spectrum. This book uses the shorter phrase only when the surrounding context makes the distinction explicit.

One of the first falsification tests for any future LSC-AW implementation is prior sensitivity. If the preference for a model disappears when the parent-energy prior is replaced by the model's own predicted spectrum, the preference was an artefact of inconsistent inference. Conversely, a model that remains viable across reasonable spectral assumptions has passed a meaningful robustness test.

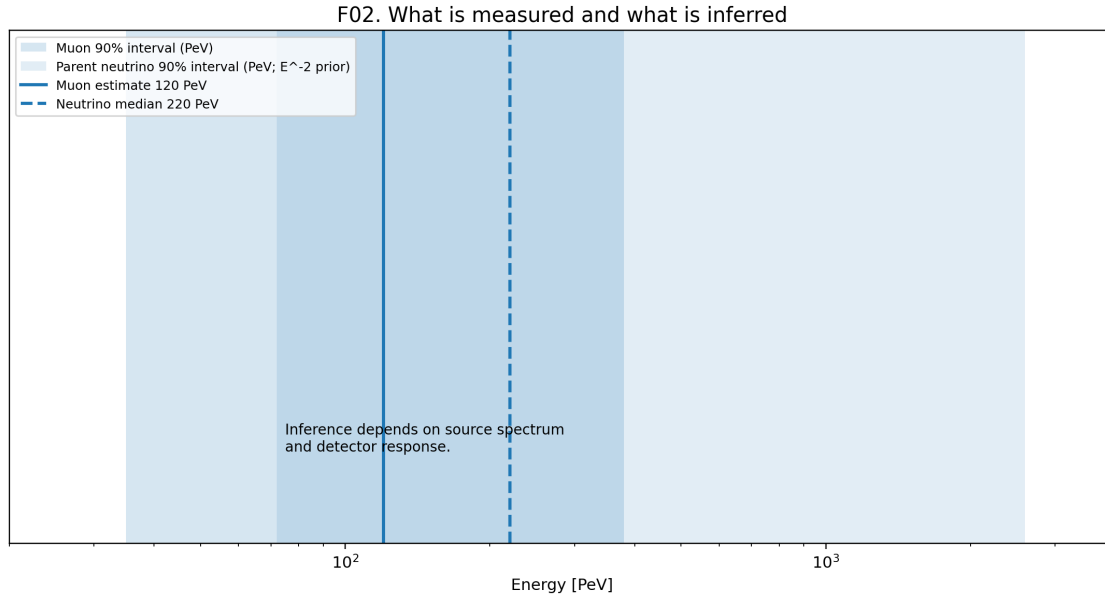


Figure 3.1: energy inference.

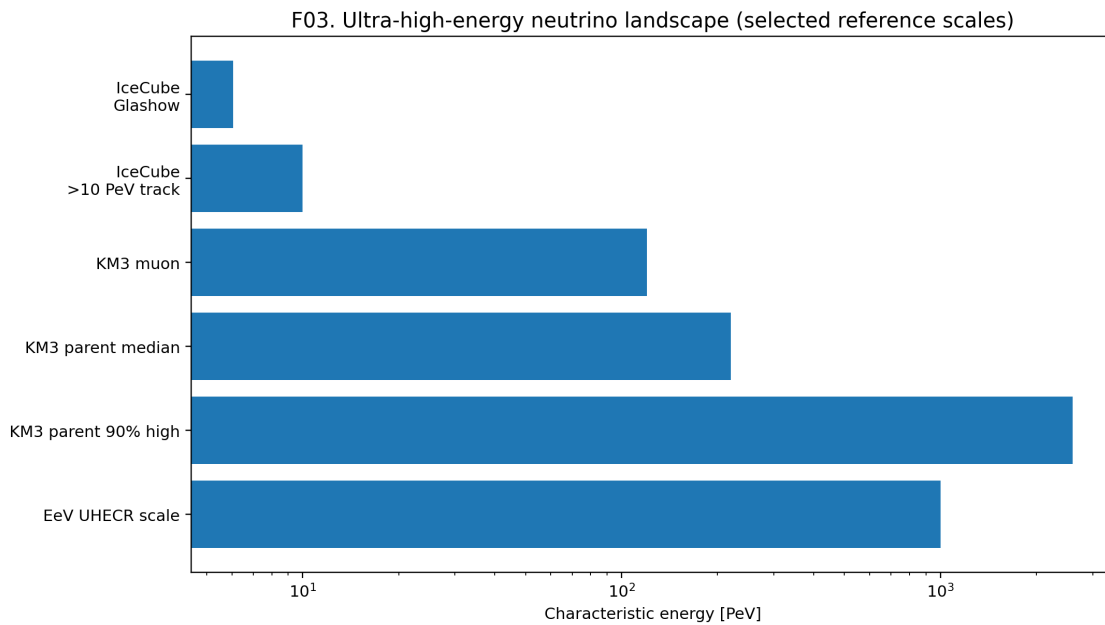


Figure 3.2: F03 energy landscape.

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The Multi-Messenger Problem

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A neutrino source rarely exists in isolation from other particles and fields. Hadronic interactions that make neutrinos generally make photons, electrons, positrons, and cosmic rays as well. Primordial-black-hole evaporation emits all kinematically available species. Heavy dark-matter decay is constrained by secondary gamma rays. Ultra-high-energy cosmic rays that generate cosmogenic neutrinos also feed electromagnetic cascades. For this reason KM3-230213A is not only a neutrino problem; it is a multi-messenger consistency problem.

The first external constraint comes from neutrino observatories themselves. IceCube has operated for much longer with a much larger instrumented volume than the partial ARCA configuration that saw the event. Pierre Auger provides complementary sensitivity to neutrinos at ultra-high energies. The absence of a comparable population in those experiments means that a steady, isotropic flux large enough to make the KM3NeT event unsurprising is disfavoured. The joint tension is moderate rather than discovery-level, of order a few standard deviations depending on spectral and statistical assumptions.

A particularly important update by September 2026 is the IceCube point-source search in the direction of KM3-230213A using fifteen years of data. The preprint reports no evidence for steady or flaring neutrino emission from that direction and sets upper limits under several temporal hypotheses. This does not eliminate every transient astrophysical model, but it narrows the space of models in which the single KM3NeT event is the bright end of a persistent source spectrum.

Electromagnetic non-detections have a subtler interpretation. No obvious gamma-ray transient was established as the counterpart. But a hadronic neutrino source can be gamma opaque: high-energy photons may pair-produce inside the source and reprocess their energy into X-rays, ultraviolet, optical, infrared, or radio emission. Thus “no gamma counterpart” is not equivalent to “no astrophysical source”. The correct question is whether a given source model predicts an electromagnetic cascade compatible with existing measurements.

The same principle is devastating for some minimal PBH scenarios. A nearby terminal Schwarzschild evaporation burst energetic enough to yield one UHE neutrino can imply enormous accompanying photon and lower-energy neutrino fluences. Published calculations using time-dependent fields of view argue that such signals should have been seen, strongly disfavoring the simplest four-dimensional scenario. Modified evaporation or hidden-sector models can alter this

conclusion, but then the modification must be quantified rather than invoked as a verbal escape hatch.

LSC-AW therefore treats non-detections as data. The joint likelihood is conceptually a product of contributions from KM3NeT, IceCube, Auger, gamma-ray observations, X-ray/radio/infrared information, and UHECR measurements. In practice, not every public dataset currently supports a full event-level likelihood. Where the information is incomplete, this book downgrades the comparison rather than inventing a missing response function.

F07. Joint multi-messenger likelihood architecture

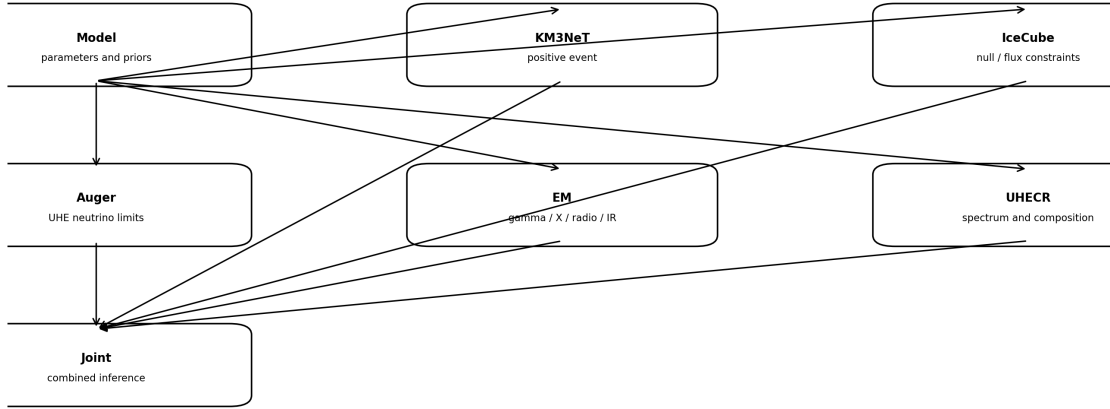


Figure 4.1: F07 joint likelihood.

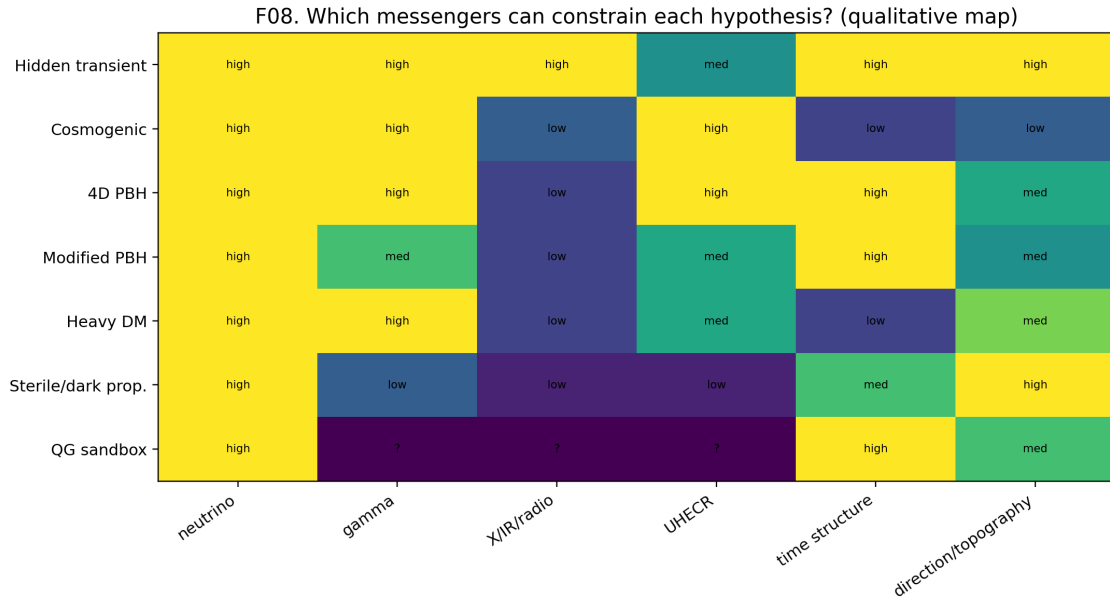


Figure 4.2: F08 multimessenger map.

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Ordinary Astrophysical Tail

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The least exotic explanation is that KM3-230213A is an unusually energetic draw from the same broad astrophysical population already established by IceCube and other neutrino telescopes. This hypothesis is scientifically indispensable because it defines the null against which more elaborate mechanisms should earn their complexity.

The published KM3NeT analysis compared the event with extrapolations of IceCube single-power-law fluxes. In the official public notebook, the expected number of bright-track ARCA events over the analyzed livetime is small: examples include about 0.117 for the IceCube Northern Sky Tracks best-fit extrapolation, 0.052 for ESTES, and 0.020 for HESE over the full energy integration used by the notebook. In the central 90 percent energy interval of KM3-230213A the expectations are even smaller. These values make the event unusual, but an unusual event is not automatically evidence for a second population.

KM3NeT quantified the standalone upward fluctuation interpretation at about 2.2 sigma for its configuration. Later global analyses that combine KM3NeT with IceCube and Auger find moderate tension, not a decisive rejection of a single astrophysical component. The important methodological point is that the null model already has uncertainty in spectral index, normalization, possible breaks or cutoffs, source evolution, and detector exposures. A fair comparison must propagate those uncertainties rather than compare an exotic model with a rigid straw-man power law.

What would strengthen the ordinary-tail explanation? Additional events distributed as expected from a smooth continuation of the diffuse spectrum, with no strong directional clustering and no new messenger correlations, would do so. What would weaken it? A repeated cluster near the same direction, a narrow energy feature, a strong time correlation with a transient class, or a detector-topography dependence would all motivate a more structured model.

The LSC-AW framework deliberately keeps AW-M1 even if later chapters focus on more dramatic physics. Any model-selection statistic must compare against this baseline with comparable nuisance freedom. A one-event fit can otherwise reward a complicated model simply because it was built around the event after the fact.

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Hidden Transient Sources

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An attractive standard-physics path is a transient source that is efficient at accelerating hadrons and producing neutrinos but is either short-lived, electromagnetically hidden, or both. Candidate classes include active galactic nuclei and blazars during flares, gamma-ray bursts, tidal-disruption events, luminous fast blue optical transients, and other powerful compact-engine transients.

The basic photohadronic channel is $p\gamma$ production. A proton of sufficiently high energy interacts with a photon field, produces pions, and charged-pion decay yields neutrinos. Hadronuclear pp interactions provide another route in sufficiently dense gas. A rough energy bookkeeping rule is that an individual neutrino carries only a few percent of the parent proton energy. A 220 PeV neutrino therefore points to a parent proton energy of order several EeV, firmly in the ultra-high-energy cosmic-ray regime.

This energy requirement makes source opacity central. The same pion production network also creates gamma rays. In a dense radiation field, however, gamma-gamma pair production can absorb TeV–EeV photons and reprocess the energy into lower bands. A neutrino-bright, gamma-dark source is therefore not paradoxical. The price is that the reprocessed energy must go somewhere, making X-ray, ultraviolet, optical, infrared, or radio constraints relevant.

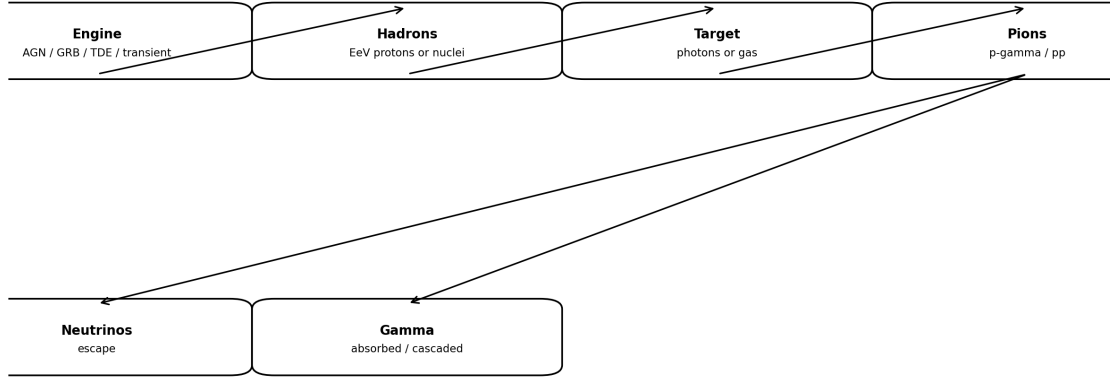
KM3NeT and follow-up studies identified multiple blazar candidates in or near the event region, but none is a compelling counterpart. A 2026 KM3NeT population study found that a population of blazars can be compatible with the joint UHE neutrino flux and Fermi-LAT diffuse gamma-ray constraints for hard proton spectra and moderate baryonic loading. The same work showed that fitting the KM3NeT event alone pushes toward much larger baryonic loading and reintroduces tension with IceCube and Auger. This is exactly the kind of result LSC-AW is intended to preserve: an astrophysical mechanism may be viable in population form while a naïve one-source normalization is disfavoured.

Long-duration GRB blast-wave models have also been used to constrain baryon loading and ambient density using the combined KM3NeT, IceCube, and Auger exposure. Such analyses are valuable because they predict an event rate, not merely a maximum energy. They also demonstrate how quickly parameter space changes when non-detections from other experiments are included.

The unique falsifiers of AW-M3 are therefore multi-dimensional: recurrence from a source position, temporal clustering, a hard but physically achievable parent spectrum, electromagnetic

reprocessing consistent with archival data, and a source population rate compatible with the rarity of the event. A hidden transient explanation remains a leading standard benchmark, but it must be more specific than the phrase “some flare somewhere”.

F10. Hidden transient source - schematic



A gamma-dark source can remain neutrino-bright if electromagnetic opacity is high.

Figure 6.1: F10 hidden transient.

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Cosmogenic Neutrinos and UHECRs

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Cosmogenic neutrinos are produced during the propagation of ultra-high-energy cosmic rays through cosmic photon backgrounds. Protons and nuclei interact with the cosmic microwave background and extragalactic background light, producing pions and unstable secondaries whose decays yield neutrinos. This mechanism has a major conceptual advantage: the neutrino does not need a bright electromagnetic counterpart at the same sky position and time as the original accelerator.

The challenge is global consistency. A cosmogenic model is tied to the injection spectrum of UHECRs, their composition, maximum rigidity, source evolution with redshift, photohadronic interaction rates, photodisintegration for nuclei, and electromagnetic cascades. The model must match the UHECR spectrum and composition measured at Earth while respecting diffuse gamma-ray limits and neutrino non-detections.

The official KM3NeT public notebook includes a set of published cosmogenic spectra. When folded through the public bright-track effective area, expected event counts span orders of magnitude. Representative full-range expectations range from roughly 10^{-3} to nearly 0.5 events for the partial ARCA exposure, depending on the model. This wide spread explains why the observation can be simultaneously surprising under many conservative models and not absurd under more optimistic ones.

Recent 2026 analyses sharpen the picture. Producing enough cosmogenic flux generally favours a hard UHE proton component and strong positive source evolution extending to substantial redshift. Yet once IceCube and Auger null observations and diffuse gamma-ray constraints are included, the strongest source evolution can become disfavoured. This is not a contradiction; it is the expected behaviour of a tightly coupled multi-messenger model.

The LSC-AW cosmogenic kernel therefore treats the event as one term in a larger population inference. It should never tune a cosmogenic spectrum only to pass through the KM3NeT flux point. The correct object is a joint fit to cosmic rays, neutrinos, and gamma rays with nuisance parameters for composition and source evolution. Until that calculation is reproduced in a common pipeline, AW-M4 remains an active published hypothesis rather than a solved origin.

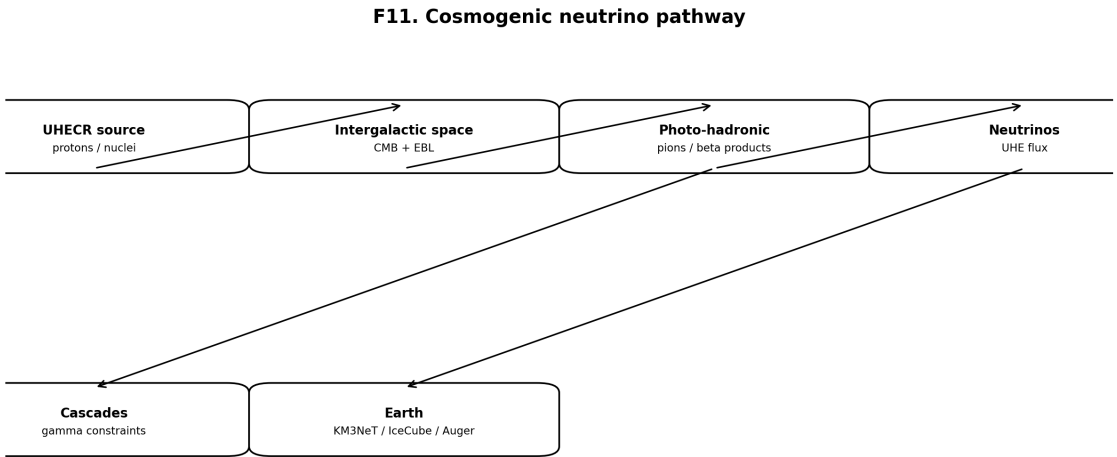


Figure 7.1: F11 cosmogenic.

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Dark Matter and Exotic Relics

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Ultra-heavy dark matter provides a qualitatively different route to UHE neutrinos. If a relic particle with mass far above the PeV scale decays over cosmological times, its decay products can include neutrinos with energies in the range relevant to KM3-230213A. The signal may contain both a Galactic halo component and an extragalactic component integrated over cosmic history.

The principal attraction is spectral reach: the energy scale can arise directly from the particle mass rather than from extreme acceleration in a conventional source. The principal difficulty is multi-messenger consistency. Most decay channels also produce electroweak and hadronic secondaries, creating gamma rays and cosmic rays that are already constrained.

A 2026 KM3NeT study used the collaboration’s detector simulation to test heavy dark-matter decay directly against the deposited-energy and arrival-direction information rather than comparing only to an E^{-2} equivalent flux. Across several two-body channels, the preferred masses under the dark-matter hypothesis lie above roughly 100 PeV at 95 percent confidence, with best-fit lifetimes around 10^{26} – 10^{27} s. The study also emphasizes that the preferred regions are in tension with other neutrino-telescope and gamma-ray bounds. This is a useful methodological advance because it illustrates how model-specific detector folding can materially change the inference relative to flux-point matching.

LSC-AW therefore gives AW-M7 an evidence status of active/preprint hypothesis rather than a favoured solution. The model has a defined energy reservoir, a calculable spectrum, and testable anisotropy. Its weakness is that the same long lifetime and large mass needed for the event can overproduce other messengers. A future population of events tracing the Galactic halo, or a characteristic spectral edge, would strengthen the case. A continued absence of the associated gamma-ray signatures would tighten the opposite conclusion.

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Sterile Neutrinos, NSI, and Detector Topography

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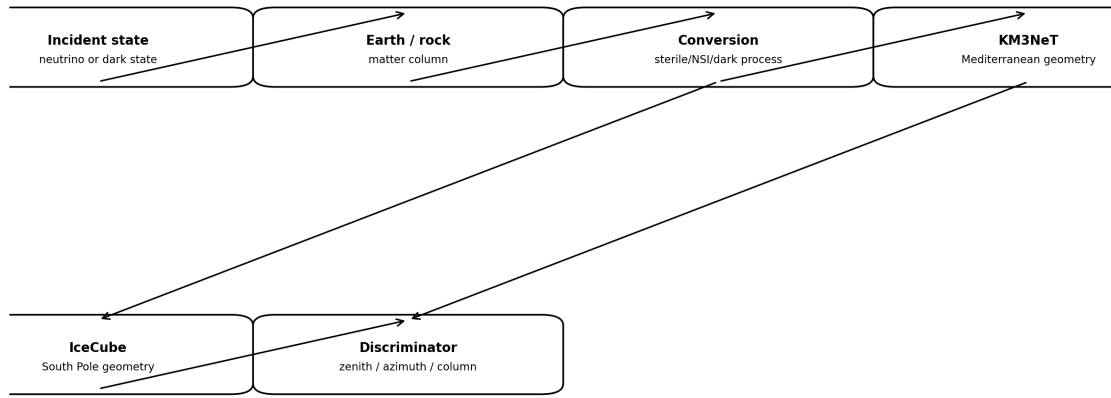
A different family of explanations does not create the UHE energy at the detector. Instead it changes which particle reaches the detector, how it propagates through matter, or how it converts near the instrument. Sterile-neutrino mixing, non-standard interactions, dark-sector particles, and metastable states have all been proposed in this broad spirit.

The distinctive clue is geometry. KM3-230213A arrived very close to the horizon, and the amount of rock or water traversed before reaching KM3NeT is different from the matter column relevant to IceCube for the same celestial direction. A new state that is weakly interacting over long distances but can convert or decay near a detector might therefore produce a signal that depends strongly on topography and zenith angle.

This class must be kept separate from the gallium sterile-neutrino problem in historical LSC. Here the sterile/NSI mechanism is an UHE propagation hypothesis. It has nothing to do with importing the LSC 6.5.0 gallium parameter. The similarity lies only in the general concept of invisible or altered neutrino-sector states.

The scientific advantage of AW-M8 is falsifiability. A topography-driven explanation should predict a distribution in zenith, azimuth, matter column, and detector site. It should also specify whether the final track is produced by a standard neutrino interaction, by a heavy metastable state decaying to a muon, or by some other process. If a model explains only one direction after the event has been observed and provides no population-level angular prediction, it has not yet earned explanatory status.

The IceCube non-detections are especially informative here because they test whether the same source flux would have produced signals under a different matter profile. Future KM3NeT data, once the detector is larger, could provide a particularly sharp test through the angular distribution of UHE candidates. LSC-AW treats that directional population prediction as the required bridge from an interesting mechanism to a viable model.

F16. Detector-topography hypothesis - schematic

Any topography explanation must predict direction-dependent rates, not only one event.

Figure 9.1: F16 topography.

What is measured? Detector-level observables and publicly documented response quantities.

What is inferred? Source or parent-particle properties that depend on model assumptions.

What would falsify it? A model-specific observation that is inconsistent with the predicted rate, spectrum, direction, companions, or causal structure.

Status of evidence. See the model cards and acceptance-gate register; absence of a calculation is labelled UNKNOWN rather than silently treated as a pass.

Primordial Black Holes: Minimal Hawking Scenario

Evidence-status rule. Throughout this book, OBSERVED means directly tied to an experimental measurement; STANDARD means established physics with ordinary astrophysical assumptions; ACTIVE-HYPOTHESIS means a published mechanism under current investigation; PREPRINT-HYPOTHESIS means relevant literature exists but peer review is incomplete; LSC-SANDBOX means an original or highly speculative extension that has not earned promotion into the active physical model set.

Primordial black holes (PBHs) can in principle emit ultra-high-energy particles near the end of Hawking evaporation. For a non-rotating uncharged black hole, the Hawking temperature is

$$T_H = \frac{\hbar c^3}{8\pi G M k_B},$$

so the temperature increases as the mass decreases. A PBH formed with an initial mass near the familiar 5×10^{14} g lifetime scale can survive until the present epoch and then enter a final phase with a much smaller instantaneous mass and a much higher temperature.

This immediately corrects a common conceptual error: a PBH with a current mass of 5×10^{14} g is not hot enough to emit 220 PeV particles efficiently. The relevant UHE emission occurs only after evaporation has reduced the remaining mass by many orders of magnitude. Published estimates place the remaining mass in the 10^5 – 10^7 g range for PeV to tens-of-PeV peak energies, with extremely short remaining lifetimes.

The supportive PBH analysis by Klipfel and Kaiser treats high-energy IceCube and KM3NeT events within a population of evaporating PBHs and derives an inferred local explosion rate from the IceCube flux. Under their assumptions, there is a non-negligible Poisson probability, about 7.6 percent, for one sufficiently close burst since 2011, corresponding to a characteristic reach of roughly 1890 AU for a large Earth-intersection area.

A critical multi-messenger analysis reaches a very different source-distance estimate when the requirement is that the PBH produce one detected KM3NeT neutrino through a detector effective area. Depending on the effective-area treatment, it finds distances of order $(1\text{--}7) \times 10^{-5}$ pc, a few astronomical units. At those distances the same Hawking burst should produce enormous photon and lower-energy neutrino signals. The predicted absence of such companions, including an especially large expected LHAASO gamma-ray signal in the hours before the event, strongly disfavors the minimal four-dimensional Schwarzschild interpretation.

These papers are not mutually reducible because they normalize the detection problem

differently. LSC-AW does not choose the friendlier result. It treats the disagreement as a reproduction target: write the source population, burst spectrum, geometric dilution, detector area, time window, and selection efficiency in a common notation, then identify exactly which assumption produces the orders-of-magnitude difference.

The status of AW-M5 in this book is therefore active hypothesis but strongly constrained. Its strengths are a clear energy reservoir and calculable emission. Its weakness is the unavoidable companion radiation of a standard Hawking burst. A minimal model that predicts hundreds or millions of missing companion events fails the multi-messenger gate even if it can generate a 220 PeV neutrino kinematically.

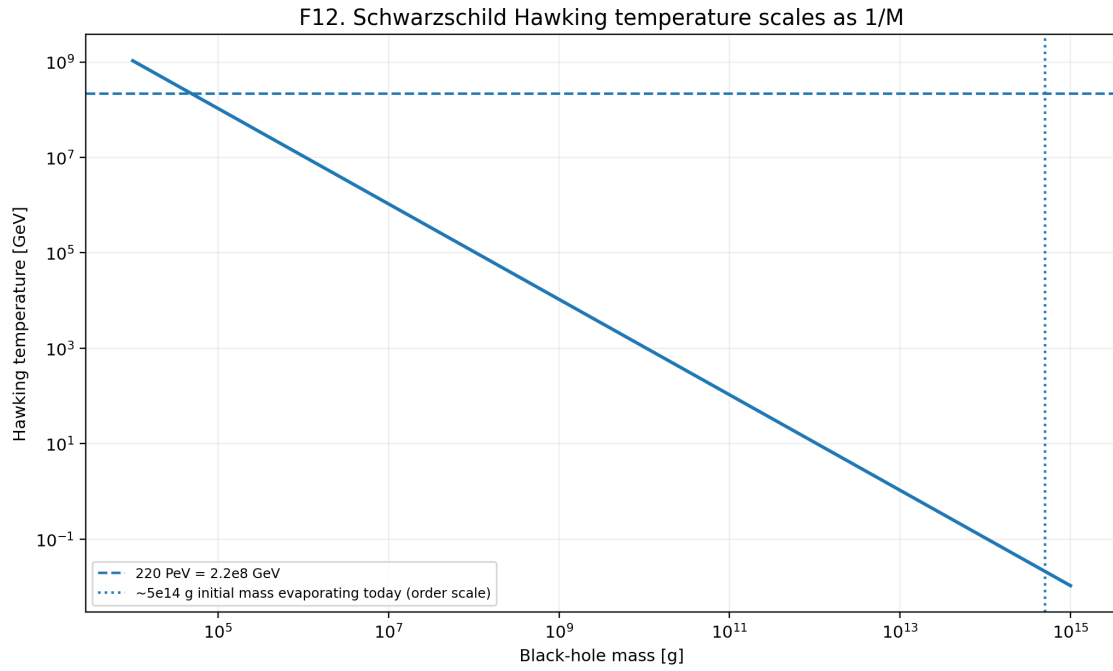


Figure 10.1: F12 pbh mass temperature.

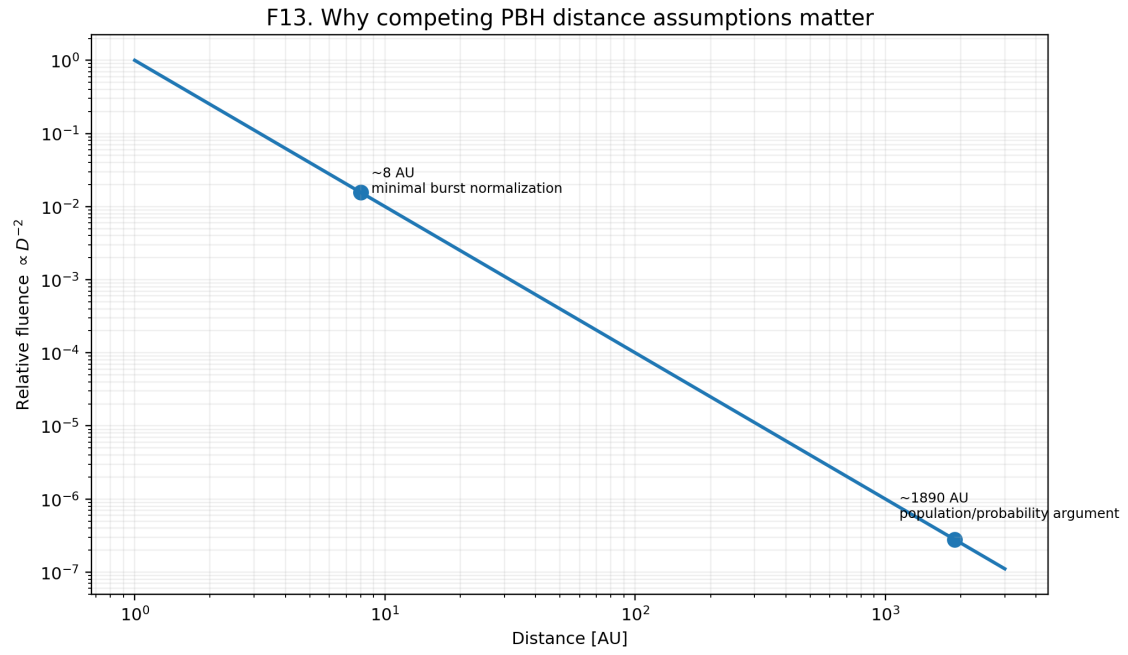


Figure 10.2: F13 pbh distance.

What is measured? Detector-level observables and publicly documented response quantities.

What is inferred? Source or parent-particle properties that depend on model assumptions.

What would falsify it? A model-specific observation that is inconsistent with the predicted rate, spectrum, direction, companions, or causal structure.

Status of evidence. See the model cards and acceptance-gate register; absence of a calculation is labelled UNKNOWN rather than silently treated as a pass.

Modified PBH Scenarios

Evidence-status rule. Throughout this book, OBSERVED means directly tied to an experimental measurement; STANDARD means established physics with ordinary astrophysical assumptions; ACTIVE-HYPOTHESIS means a published mechanism under current investigation; PREPRINT-HYPOTHESIS means relevant literature exists but peer review is incomplete; LSC-SANDBOX means an original or highly speculative extension that has not earned promotion into the active physical model set.

The failure of the minimal Schwarzschild burst does not automatically eliminate every PBH-related explanation. Several published extensions alter the late-time evolution, the particle content, or the spacetime geometry. Examples include memory-burden scenarios, quasi-extremal PBHs carrying a hidden charge, higher-dimensional black holes, and regular or black-bounce metrics.

The scientific burden rises with each modification. A new degree of freedom is not a free escape from a failed constraint; it must have its own prior, dynamics, emission spectrum, event rate, and independent consequences. The right comparison is not “standard PBH fails, therefore modified PBH works.” It is “what exact modification suppresses the excluded companions while preserving enough UHE neutrino signal, and what new observable does that modification create?”

BlackHawk v3.0 is important because it extends a public Hawking-radiation code beyond Schwarzschild and Kerr-type geometries to several regular and black-bounce metrics, including Bardeen, Hayward, Simpson-Visser, Peltola-Kunstatter, and a D’Ambrosio-Rovelli black-hole-to-white-hole geometry. The release computes Hawking temperatures and greybody factors for those metrics and therefore provides a concrete route to test how spectral hardness and emission rates change when the singular interior is regularized.

There is an important limitation, however. BlackHawk v3.0 can compute instantaneous spectra for these geometries, but a fully consistent evaporation history may require a dynamical prescription for how the regularizing parameter evolves. A static metric plus an instantaneous spectrum is not yet a complete burst model. LSC-AW therefore separates “geometry supported by code” from “time-dependent population prediction.”

Quasi-extremal and extra-dimensional models can reduce visible electromagnetic output or redirect energy into hidden states. Such models are legitimate active hypotheses when their equations are explicit. They remain constrained by abundance, lifetime, cosmology, and any visible decay products. A successful AW-M6 model must pass the same cross-experiment and rate gates as AW-M5; it is not exempt because its microphysics is more elaborate.

The book therefore treats modified PBHs as a family of model cards rather than a single bin. Each card states the metric or field content, the energy reservoir, the emission channels, the quantity that replaces the Schwarzschild temperature law if needed, and the unique observation that could falsify it.

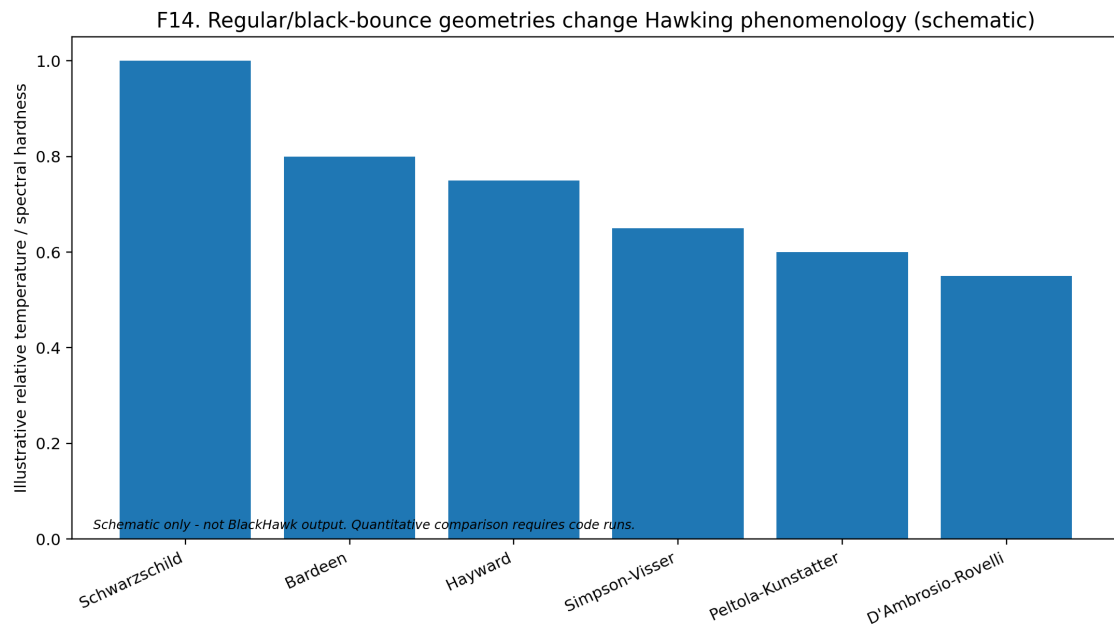


Figure 11.1: F14 geometry schematic.

What is measured? Detector-level observables and publicly documented response quantities.

What is inferred? Source or parent-particle properties that depend on model assumptions.

What would falsify it? A model-specific observation that is inconsistent with the predicted rate, spectrum, direction, companions, or causal structure.

Status of evidence. See the model cards and acceptance-gate register; absence of a calculation is labelled UNKNOWN rather than silently treated as a pass.

LSC Lineage and the Boundary of the New Programme

Evidence-status rule. Throughout this book, OBSERVED means directly tied to an experimental measurement; STANDARD means established physics with ordinary astrophysical assumptions; ACTIVE-HYPOTHESIS means a published mechanism under current investigation; PREPRINT-HYPOTHESIS means relevant literature exists but peer review is incomplete; LSC-SANDBOX means an original or highly speculative extension that has not earned promotion into the active physical model set.

The name LSC predates the KM3NeT event and has a separate experimental lineage in the gallium-neutrino anomaly. That history matters because it contains useful methodological lessons, but the two physical problems must not be conflated.

The September 2026 LSC research book records several version boundaries. LSC 6.3.0 preserved a phenomenological gallium framework with explicit comparison baselines and a pre-BEST-2 freeze philosophy. LSC 6.4.0 produced an adverse development result dominated by the boundary of a permitted linear approximation; the record explicitly says this should not be presented as a successful model, while also cautioning that a development-gate failure is not automatically an empirical falsification of every historical LSC idea. LSC 6.5.0 is the cleanest executable object: an exact finite rescaling of the argument used in the gallium capture cross section. Its status remains NOT_YET_VALIDATED.

The 6.5.0 equations are

$$E'_i = E_i \exp(\alpha_0), \quad K_i = \frac{\sigma(E'_i)}{\sigma(E_i)},$$

$$R = \frac{\sum_i q_i \sigma(E'_i)}{\sum_i q_i \sigma(E_i)}.$$

The frozen parameter is about $\alpha_0 = -0.1438879552$, so $\exp(\alpha_0) \simeq 0.865985$. In that model only the argument of the gallium capture-response function is shifted. The source energy, propagation probability, geometry, and time are not changed. The historical audit explicitly warns that the 13.4 percent scale difference must not be re-described as a measured neutrino energy loss.

This matters for the present book. Nothing in KM3-230213A justifies carrying α_0 into UHE physics. The gallium model and the UHE event involve different energies, interactions, detectors, and observables. Any cross-sector connection would require an independent derivation from a microscopic theory and would then be a new model version.

What LSC-AW inherits is not a parameter. It inherits a discipline: separate observation from

inference, keep competing baselines, preserve adverse results, freeze predictions before new data, and do not call a calibrated fit a validation. LSC-AW is therefore a post-freeze research lineage. Its core is an inference architecture for the UHE neutrino event, not the 6.5.0 gallium response law.

What is measured? Detector-level observables and publicly documented response quantities.

What is inferred? Source or parent-particle properties that depend on model assumptions.

What would falsify it? A model-specific observation that is inconsistent with the predicted rate, spectrum, direction, companions, or causal structure.

Status of evidence. See the model cards and acceptance-gate register; absence of a calculation is labelled UNKNOWN rather than silently treated as a pass.

LSC-AW Core

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The strongest defensible form of Alternative Wave is deliberately model independent. Introduce an additional flux component

$$\Delta\Phi_{\text{AW},\alpha}(E, \Omega, t) = A_\alpha f_E(E; E_\star, \sigma_E) f_\Omega(\Omega; \hat{n}_\star, \sigma_\Omega) f_t(t; t_\star, \tau).$$

This equation does not say what the component is. It merely gives the data a language for asking whether the anomaly prefers a particular energy scale, angular localization, time scale, or flavour content beyond the ordinary baseline.

For a microscopic model M , the expected event count in detector d and analysis class k is written schematically as

$$\mu_{d,k}(\Theta_M) = \sum_{\alpha,\beta} \int dE d\Omega dt \Phi_{\alpha,M} P_{\alpha \rightarrow \beta, M} A_{\text{eff},\beta}^{(d)} R_k^{(d)}.$$

The separation is the point. Production belongs in Φ . Propagation or conversion belongs in P . Detector sensitivity belongs in A_{eff} . Reconstruction belongs in R . A model that changes two layers must say so explicitly.

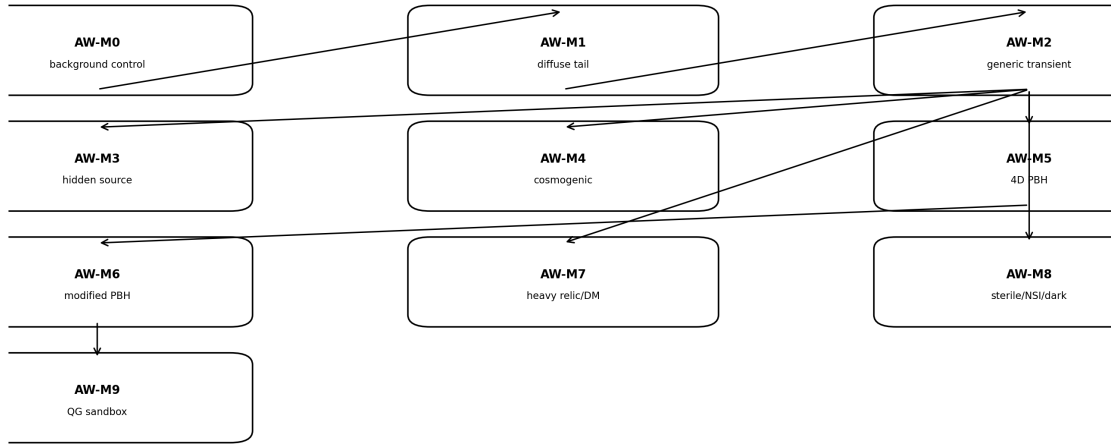
This architecture protects against a common failure in anomaly-driven theory building: moving an unexplained discrepancy from one layer to another without noticing. For example, a dark-particle model may fit a track energy by changing propagation, while a hidden transient changes production, and a detector-systematics explanation changes response. If all are encoded as “an extra flux”, they look artificially similar. The layer decomposition keeps their causal claims distinct.

The model hierarchy AW-M0 through AW-M9 is therefore a registry, not a ladder of plausibility. AW-M0 and AW-M1 are controls. AW-M2 is a generic transient component. AW-M3 and M4 are conventional astrophysical and cosmogenic models. AW-M5 and M6 cover minimal and modified PBHs. AW-M7 covers heavy relics. AW-M8 covers sterile/NSI/dark propagation. AW-M9 is reserved for quantum geometry and remains sandbox unless it produces a normalized spectrum, a rate, a causal escape mechanism, and a unique falsifier.

The LSC contribution is methodological: force all these models through the same gates. The

result may be that conventional astrophysics wins, that several models remain indistinguishable, or that a new component is required. Any of those outcomes is scientifically useful.

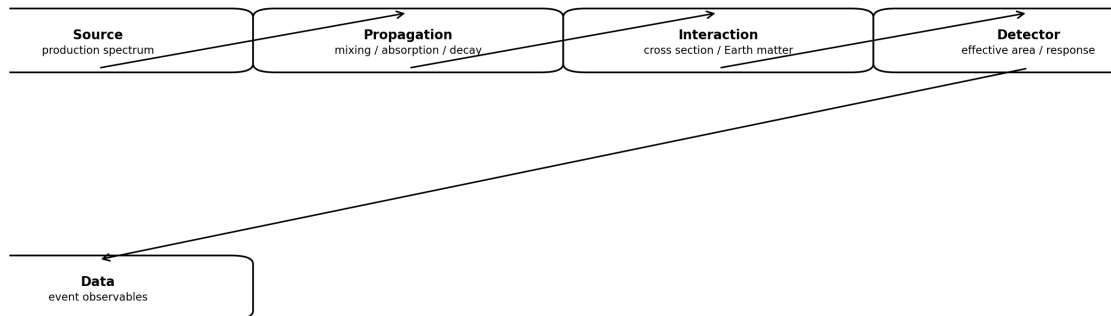
F05. LSC-AW model hierarchy



Hierarchy is an analysis registry, not a ranking of truth.

Figure 13.1: F05 model hierarchy.

F06. Source - propagation - detector causal separation



Each model must state which layer it changes.

Figure 13.2: F06 dag.

What is measured? Detector-level observables and publicly documented response quantities.

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What would falsify it? A model-specific observation that is inconsistent with the predicted rate, spectrum, direction, companions, or causal structure.

Status of evidence. See the model cards and acceptance-gate register; absence of a calculation is labelled UNKNOWN rather than silently treated as a pass.

Joint Likelihood and Statistical Language

Evidence-status rule. Throughout this book, OBSERVED means directly tied to an experimental measurement; STANDARD means established physics with ordinary astrophysical assumptions; ACTIVE-HYPOTHESIS means a published mechanism under current investigation; PREPRINT-HYPOTHESIS means relevant literature exists but peer review is incomplete; LSC-SANDBOX means an original or highly speculative extension that has not earned promotion into the active physical model set.

The statistical architecture must prevent a model from being rewarded for explaining the one positive event while ignoring the many relevant non-detections. A schematic joint likelihood is

$$\mathcal{L}_{\text{joint}}(M) = \mathcal{L}_{\text{KM3NeT}} \mathcal{L}_{\text{IceCube}} \mathcal{L}_{\text{Auger}} \mathcal{L}_{\gamma} \mathcal{L}_{\text{radio}} \mathcal{L}_{\text{IR}} \mathcal{L}_X \mathcal{L}_{\text{UHECR}}.$$

Where event-level information is available, a point-process likelihood is preferable:

$$\mathcal{L}_{d,M} = e^{-\mu_{d,M}} \prod_{i=1}^{N_d} \int dE d\Omega dt \lambda_{d,M}(E, \Omega, t) p_d(y_i | E, \Omega, t).$$

The word “joint” should not be ceremonial. Units, livetime, flavour conventions, neutrino/antineutrino averaging, sky averaging, and selection definitions must be harmonized. A single effective area may be appropriate for a diffuse flux and wrong for a minute-scale point-source burst. A sky-averaged exposure may be sufficient for one comparison and unacceptable for another. Those differences are model assumptions and belong in the likelihood, not in a footnote after the result.

Bayesian evidence is useful only if priors are explicit. A Bayes factor does not create objectivity if a model’s energy scale, width, source duration, and population rate are assigned arbitrary priors after seeing the event. Profile likelihood can reduce prior dependence but still requires a defined parameter space and treatment of nuisance parameters. Posterior predictive checks are especially valuable because they ask whether replicated data from the fitted model resemble what was actually observed.

Trials factors matter because the event triggered searches across many catalogues, time windows, source classes, and exotic mechanisms. A low pre-trial probability for a source association can become unremarkable after accounting for the number of sources and tests. The published multi-detector follow-up illustrates this directly: the largest IceCube excess near the event direction had a very small pre-trial p-value but a post-trial p-value around 0.07, not significant.

The book therefore avoids numerical “probability that LSC is true” statements. When full evidences are unavailable, it reports likelihood ratios, confidence regions, predicted counts, qualitative evidence status, and explicit blockers. Scientific uncertainty is not a defect to be hidden; it is the information that tells the next experiment what to measure.

What is measured? Detector-level observables and publicly documented response quantities.

What is inferred? Source or parent-particle properties that depend on model assumptions.

What would falsify it? A model-specific observation that is inconsistent with the predicted rate, spectrum, direction, companions, or causal structure.

Status of evidence. See the model cards and acceptance-gate register; absence of a calculation is labelled UNKNOWN rather than silently treated as a pass.

Sterile/NSI versus LSC-AW

Evidence-status rule. Throughout this book, OBSERVED means directly tied to an experimental measurement; STANDARD means established physics with ordinary astrophysical assumptions; ACTIVE-HYPOTHESIS means a published mechanism under current investigation; PREPRINT-HYPOTHESIS means relevant literature exists but peer review is incomplete; LSC-SANDBOX means an original or highly speculative extension that has not earned promotion into the active physical model set.

Sterile-neutrino or NSI models and LSC-AW Core are not the same kind of object. A sterile/NSI model is a microscopic or effective physical hypothesis that modifies propagation or interaction. LSC-AW Core is initially an inference architecture designed to compare such hypotheses. Calling one the “winner” over the other without specifying a concrete LSC microscopic model would be a category error.

Nevertheless, the comparison is useful. Sterile/NSI models usually predict explicit dependence on energy, matter density, baseline, or mixing parameters. Their advantage is reach: the same parameters can be constrained by many experiments. Their disadvantage is exactly the same: because they make broad predictions, they are already strongly restricted by external data. A detector-topography model can be especially predictive because KM3NeT and IceCube see the same sky through different matter columns.

LSC-AW at architecture level makes fewer physical claims. It can remain compatible with more observations simply because it is not yet a universal particle theory. That is not evidence in its favour. To become a competitor rather than a bookkeeping system, a specific LSC-AW extension must define a production or propagation mechanism, parameters, priors, event rate, and external predictions.

The discriminating observables are therefore more important than raw fit quality. If future UHE events cluster by matter column rather than by source population, AW-M8 gains support. If events track transient astrophysical populations with electromagnetic reprocessing, AW-M3 gains support. If an isotropic spectrum follows UHECR source evolution, AW-M4 gains support. If a new LSC-specific mechanism predicts a pattern not shared by these competitors, that prediction can be frozen and tested.

The historical gallium LSC 6.5.0 model does not enter this contest. It modifies a low-energy gallium capture response and contains no UHE production or propagation law. Its only legitimate role here is methodological lineage.

What is measured? Detector-level observables and publicly documented response quantities.

What is inferred? Source or parent-particle properties that depend on model assumptions.

What would falsify it? A model-specific observation that is inconsistent with the predicted

rate, spectrum, direction, companions, or causal structure.

Status of evidence. See the model cards and acceptance-gate register; absence of a calculation is labelled UNKNOWN rather than silently treated as a pass.

Standard Astrophysics versus PBH versus LSC-AW

Evidence-status rule. Throughout this book, OBSERVED means directly tied to an experimental measurement; STANDARD means established physics with ordinary astrophysical assumptions; ACTIVE-HYPOTHESIS means a published mechanism under current investigation; PREPRINT-HYPOTHESIS means relevant literature exists but peer review is incomplete; LSC-SANDBOX means an original or highly speculative extension that has not earned promotion into the active physical model set.

The most informative comparison is organized around causal obligations rather than narrative appeal. Hidden transients and cosmogenic models use established particle physics but require extreme source conditions. PBH models provide a compact energy reservoir and a calculable Hawking spectrum but struggle with companion signals and population rates. Heavy dark matter has natural access to enormous energies but faces gamma-ray and lifetime bounds. Sterile/dark propagation models can exploit detector geometry but must explain the parent UHE flux or introduce a new source. Quantum-geometry ideas may change causal structure but currently lack normalized event predictions.

LSC-AW Core does not compete on the same axis. It defines the common arena in which these models are scored. Its value is highest precisely when no single hypothesis is yet selected by the data. That also means the architecture cannot claim success by surviving constraints that were never mapped onto a physical LSC mechanism.

A practical model-comparison table therefore includes the energy reservoir, production channel, propagation law, detector dependence, predicted companion messengers, event-rate capability, parameter count, strongest current constraint, and unique falsifier. Qualitative maturity is recorded separately from fit quality.

At the present evidence level, standard hidden-transient and cosmogenic scenarios remain essential benchmarks. Minimal 4D PBH evaporation is under strong multi-messenger pressure. Modified PBHs remain possible only model by model. Heavy dark matter is actively testable but constrained. Sterile/NSI/dark topography models are interesting because they make detector-dependent predictions. Quantum bounce models are not yet event explanations because a geometry alone is not a rate calculation.

The key conclusion is intentionally modest: KM3-230213A has opened an energy regime in which multiple mechanisms are scientifically discussable, but the data do not yet justify collapsing the hierarchy to one origin. The correct response is a sharper inference framework and more prospective predictions, not stronger adjectives.

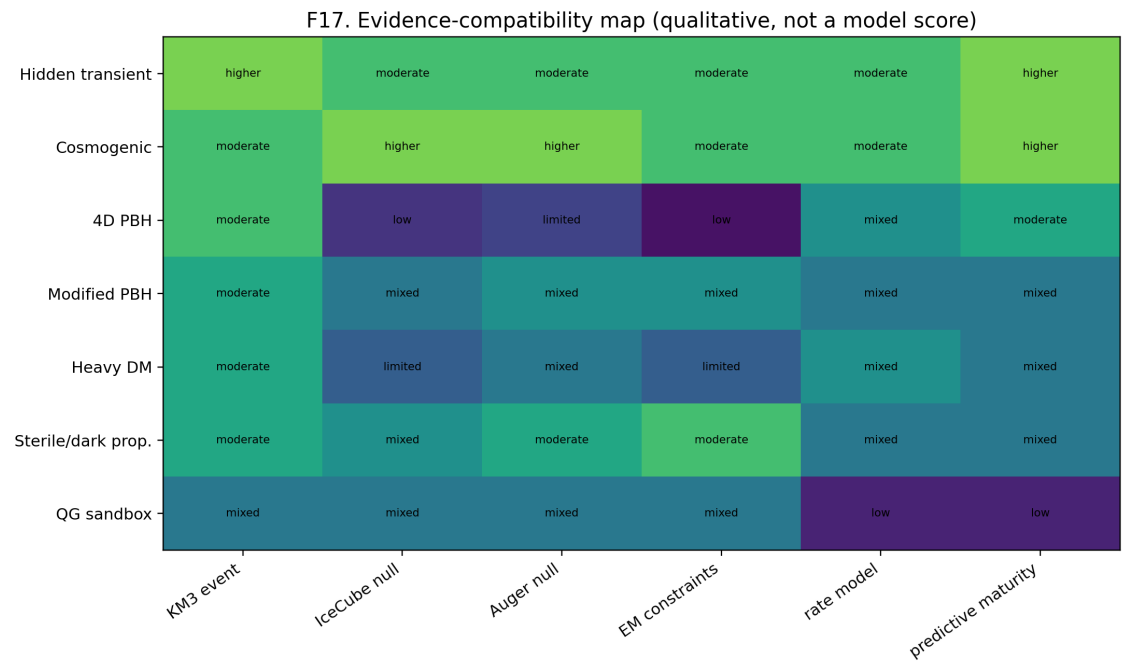


Figure 16.1: F17 evidence heatmap.

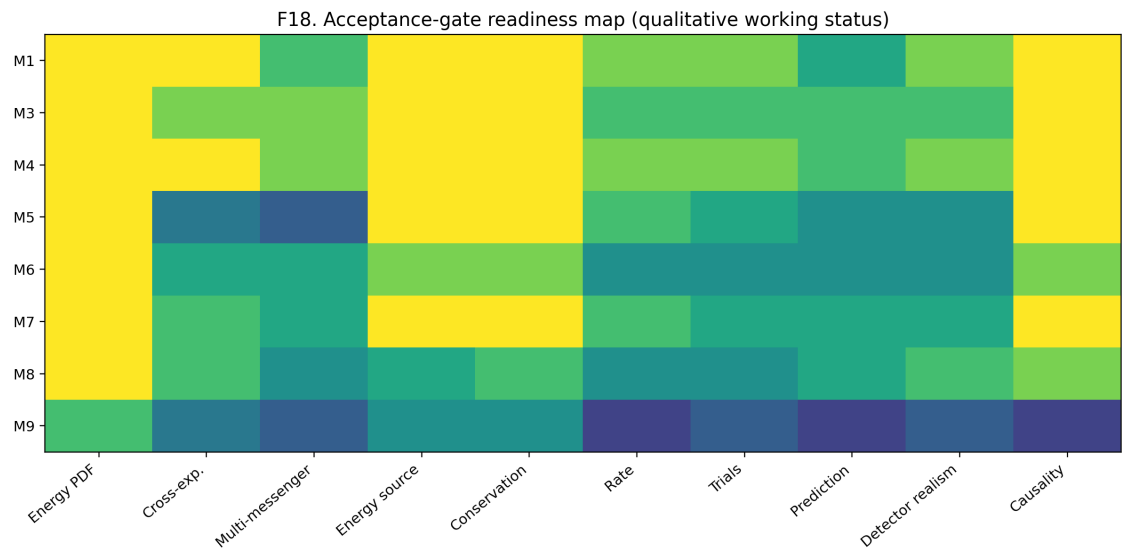


Figure 16.2: F18 gate matrix.

What is measured? Detector-level observables and publicly documented response quantities.
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Status of evidence. See the model cards and acceptance-gate register; absence of a calculation is labelled UNKNOWN rather than silently treated as a pass.

Quantum Geometry Sandbox

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The original intuitive idea behind the most speculative branch was that extreme gravitational collapse might act as an energy-conversion environment and that a particle could emerge from a non-classical interior with much higher energy. In classical general relativity this picture fails if it assumes a permanent event horizon: future-directed causal trajectories that cross such a horizon do not return to the exterior.

A quantum-gravity branch must therefore modify the global causal structure. Published black-to-white-hole and black-bounce geometries demonstrate that mathematically explicit non-singular alternatives can be constructed. BlackHawk v3.0 even includes one black-hole-to-white-hole geometry for Hawking-spectrum calculations. But existence of a metric is not evidence that KM3-230213A experienced that geometry.

The minimum LSC-AW sandbox object is a transition kernel

$$K_{QG}(E_{\text{out}}, f_{\text{out}}, \Omega_{\text{out}}, \Delta t \mid E_{\text{in}}, f_{\text{in}}; M, a, \lambda_{QG}).$$

It must be normalized and must specify the outgoing energy, flavour, angle, delay, absorption probability, and source-object state change. Energy accounting requires

$$E_{\text{in}} + M_i c^2 = E_{\text{out}} + M_f c^2 + E_{\text{other}}.$$

If the neutrino gains energy, the source or geometry must lose it. “Quantum foam” cannot be treated as an unlimited energy reservoir.

The model must also predict a rate. A schematic rate integral includes the PBH mass distribution, incident flux, interaction or capture probability, transition probability, escape probability, and detector effective area. Without a rate, the mechanism can at most show kinematic possibility.

Several historical notions are archived rather than promoted: tachyonic escape, literal information cloning, arbitrary vacuum-energy extraction, and an unspecified traversable wormhole. The sandbox is not a trash bin for unfalsifiable stories. It is a place where speculative ideas are allowed only if the work required to make them scientific is stated explicitly.

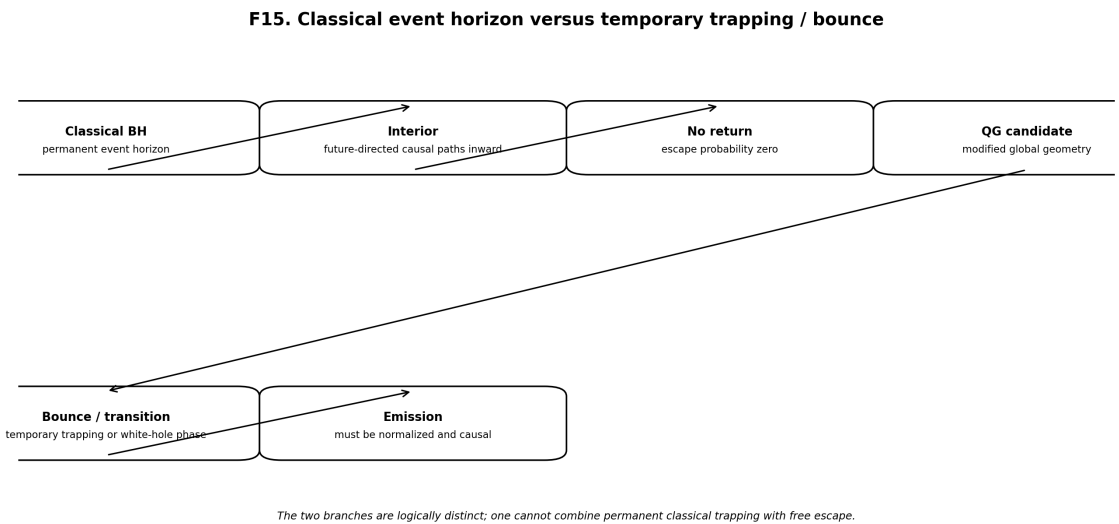


Figure 17.1: F15 causal.

What is measured? Detector-level observables and publicly documented response quantities.

What is inferred? Source or parent-particle properties that depend on model assumptions.

What would falsify it? A model-specific observation that is inconsistent with the predicted rate, spectrum, direction, companions, or causal structure.

Status of evidence. See the model cards and acceptance-gate register; absence of a calculation is labelled UNKNOWN rather than silently treated as a pass.

What Would the Next Event Look Like?

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Retrospective explanation is easy because the event is already known. Prospective prediction is harder and scientifically more valuable. The LSC-AW programme therefore requires every surviving model to produce a frozen prediction package before the next relevant event is examined.

The package should include an energy distribution $p(E_{\text{next}} | M)$, a directional distribution $p(\Omega_{\text{next}} | M)$, a waiting-time or recurrence distribution $p(\Delta t_{\text{next}} | M)$, expected relative counts in KM3NeT, IceCube, and Auger, and the probability or fluence of electromagnetic companions. A model that cannot predict one of these quantities must state why.

The predictions differ qualitatively. A diffuse cosmogenic component should produce broad sky coverage tied to UHECR source evolution, not repeated flares from one direction. A hidden transient source population should produce temporal clustering around rare source episodes and may show multiwavelength reprocessing. A Galactic heavy-dark-matter contribution should have a halo anisotropy. A detector-topography mechanism should correlate with matter column and detector site. A PBH burst should have a characteristic rapid time evolution and, in minimal scenarios, copious companions.

AW-M9 currently cannot provide a defensible normalized next-event distribution because no accepted transition kernel and population rate have been specified. Its prediction status therefore remains blocked. This is not a reason to hide the model; it is the correct status label.

A frozen prediction is stored with a model version, a timestamp, and a hash. When new data arrive, the old prediction is evaluated before any refit. If it fails, the failure remains part of the record even if a new model version is later motivated. This is the strongest methodological inheritance from the earlier LSC programme.

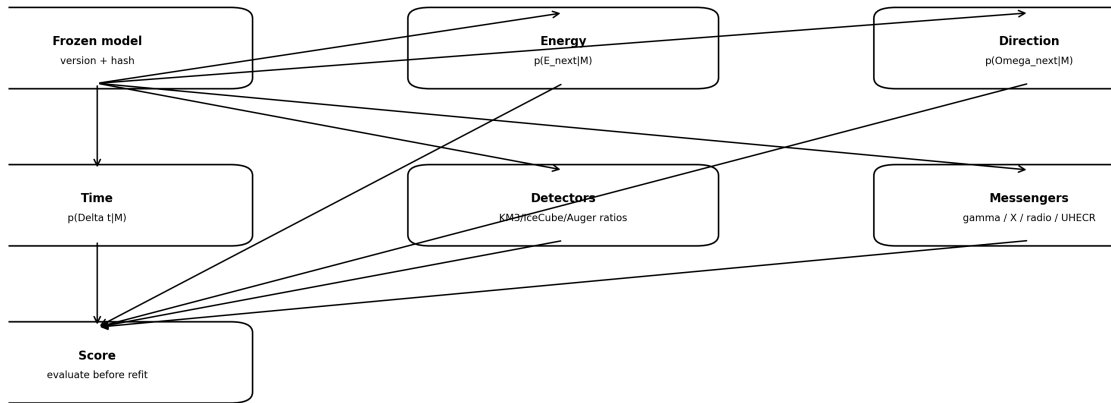
F19. Prospective prediction package

Figure 18.1: F19 predictions.

What is measured? Detector-level observables and publicly documented response quantities.

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What would falsify it? A model-specific observation that is inconsistent with the predicted rate, spectrum, direction, companions, or causal structure.

Status of evidence. See the model cards and acceptance-gate register; absence of a calculation is labelled UNKNOWN rather than silently treated as a pass.

What Would Change Our Mind?

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A scientific model should come with conditions under which its authors would reduce confidence in it. LSC-AW makes those conditions explicit.

For AW-M1, repeated UHE events concentrated in one direction or tied to transient activity would weaken the simple diffuse-tail interpretation. For AW-M3, the absence of any plausible source population after substantially improved localization and deep multiwavelength coverage would weaken hidden-transient models; conversely, repeated neutrinos from an active source with compatible gamma opacity would strengthen them. For AW-M4, UHECR composition and gamma-ray data that exclude the required proton fraction or source evolution would be decisive.

For AW-M5, the current absence of predicted companion photons and lower-energy neutrinos already constitutes a serious challenge. A future nearby burst with the expected Hawking time sequence would be transformative; continued null companion searches make the minimal scenario progressively less credible. AW-M6 can survive only if its modification both suppresses excluded signals and introduces a new prediction that can be tested independently.

For AW-M7, improved gamma-ray limits and neutrino anisotropy tests directly challenge the heavy-dark-matter parameter space. For AW-M8, an angular distribution inconsistent with matter-column dependence would weaken topography-based conversion models. For AW-M9, failure to construct a normalized causal kernel is itself a development failure: the model should remain sandbox regardless of how intuitively attractive it seems.

Finally, what would change our mind about LSC-AW as an architecture? If a single mature physical model becomes decisively established, the broad hierarchy can be simplified. Until then, the architecture earns its keep by preventing evidence from being silently moved between layers and by recording exactly which observation could discriminate each branch.

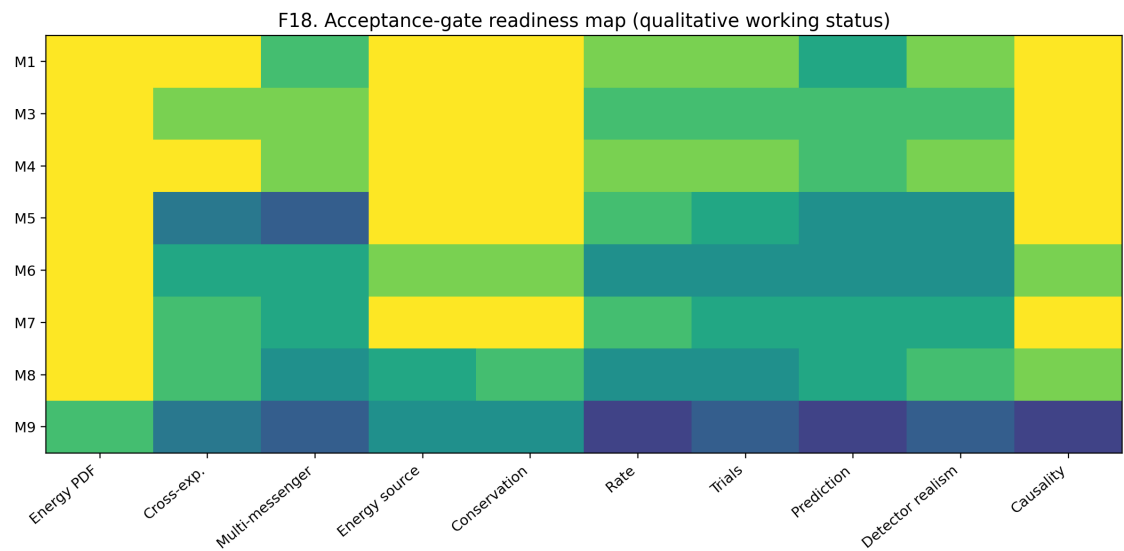


Figure 19.1: F18 gate matrix.

What is measured? Detector-level observables and publicly documented response quantities.
What is inferred? Source or parent-particle properties that depend on model assumptions.
What would falsify it? A model-specific observation that is inconsistent with the predicted rate, spectrum, direction, companions, or causal structure.
Status of evidence. See the model cards and acceptance-gate register; absence of a calculation is labelled UNKNOWN rather than silently treated as a pass.

From Retrospective Flexibility to Prospective Science

Evidence-status rule. Throughout this book, OBSERVED means directly tied to an experimental measurement; STANDARD means established physics with ordinary astrophysical assumptions; ACTIVE-HYPOTHESIS means a published mechanism under current investigation; PREPRINT-HYPOTHESIS means relevant literature exists but peer review is incomplete; LSC-SANDBOX means an original or highly speculative extension that has not earned promotion into the active physical model set.

Anomaly research is vulnerable to a particular form of overfitting: every new observation inspires a new mechanism, and every failed mechanism is modified after the fact. LSC-AW counters this with model versioning and data classification.

New data are classified as known-development, unseen-retrospective, prospective-frozen, or blind-protected. Data already known when a model was designed may be used for development but not called validation. Historical data not previously inspected can be a useful retrospective test if independence is documented. Genuine prospective validation requires a prediction frozen before the result becomes available. Blind-protected data must remain unseen until the agreed unblinding step.

When a new dataset appears, the current frozen model is run against it before any equations or priors are changed. The result is recorded under the old version. Only then may a new branch be created if there is a scientific reason to change the model. The data that motivated the change cannot also validate the changed model.

This is especially important for the KM3NeT literature because the field is moving rapidly. New point-source searches, updated effective areas, improved event localization, or additional UHE candidates can materially alter the comparison. A living book may update its literature and its model cards, but it should not rewrite yesterday's prediction into tomorrow's success.

The same rule protects the historical gallium programme. BEST-2 is treated as FUTURE_BLIND in the preserved LSC record until a formal unblinding decision exists. The present UHE programme does not inspect or use that result to tune anything in LSC-AW.

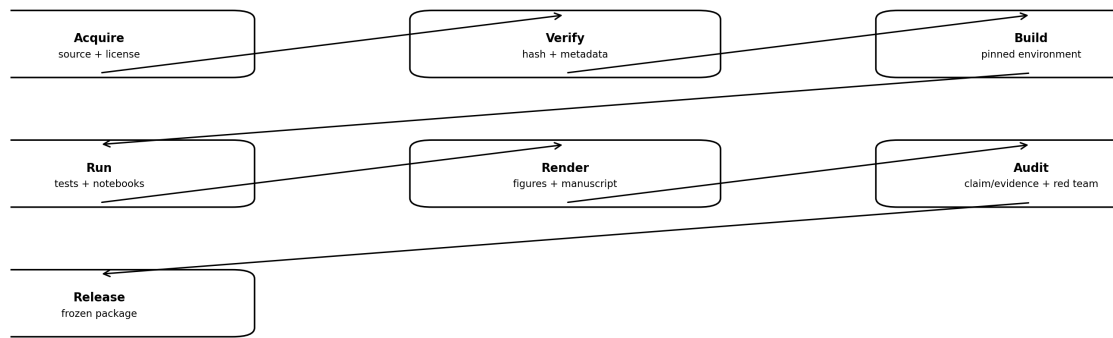
F23. Reproducibility workflow

Figure 20.1: F23 repro.

What is measured? Detector-level observables and publicly documented response quantities.

What is inferred? Source or parent-particle properties that depend on model assumptions.

What would falsify it? A model-specific observation that is inconsistent with the predicted rate, spectrum, direction, companions, or causal structure.

Status of evidence. See the model cards and acceptance-gate register; absence of a calculation is labelled UNKNOWN rather than silently treated as a pass.

Reproducibility, Data, and Code

Evidence-status rule. Throughout this book, OBSERVED means directly tied to an experimental measurement; STANDARD means established physics with ordinary astrophysical assumptions; ACTIVE-HYPOTHESIS means a published mechanism under current investigation; PREPRINT-HYPOTHESIS means relevant literature exists but peer review is incomplete; LSC-SANDBOX means an original or highly speculative extension that has not earned promotion into the active physical model set.

A review manuscript about one extraordinary event can become narrative-heavy very quickly. Reproducibility is the counterweight. Every quantitative statement should trace to a primary source or to code that can be rerun from versioned inputs.

The official KM3NeT public repository provides a strong starting point. It includes the bright-track effective area, source data for the event sky map and hit distributions, flux constraints, model spectra, and notebooks for selected paper outputs. The public effective-area file used in this build is identified by the Git object SHA recorded in the repository. The expected-events notebook defines the livetime and integration formula. Those public artifacts are preferable to hand-digitizing plots.

A complete LSC-AW package should separate immutable raw inputs from processed products, pin the software environment, record checksums, and regenerate figures from code. Figma or Canva may improve typography and composition but must never manually move data points, error bars, or confidence bands. A visual asset without a provenance path back to data and code is not a quantitative scientific figure.

This book also distinguishes reproduction from validation. Reproducing a published table checks that the computational pathway has been understood. It does not establish that the underlying model is true. Similarly, a successful unit test checks software behaviour, not physics.

The clean-room rebuild criterion is straightforward: an independent researcher should be able to obtain the cited public inputs, verify hashes, install the pinned environment, run the notebooks, regenerate the figures, build the manuscript, and understand which claims are observations, inferences, hypotheses, or sandbox proposals. Anything that depends on private memory or an unrecorded manual step is not yet release-grade.

F22. Data/code provenance graph

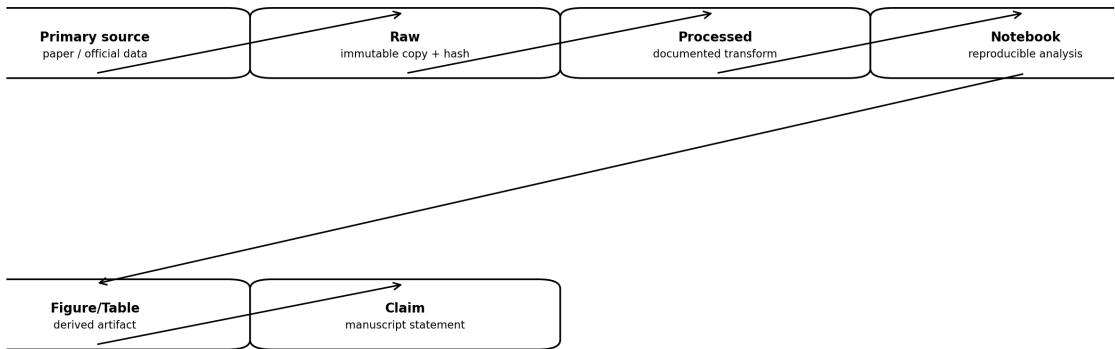


Figure 21.1: F22 provenance.

F23. Reproducibility workflow

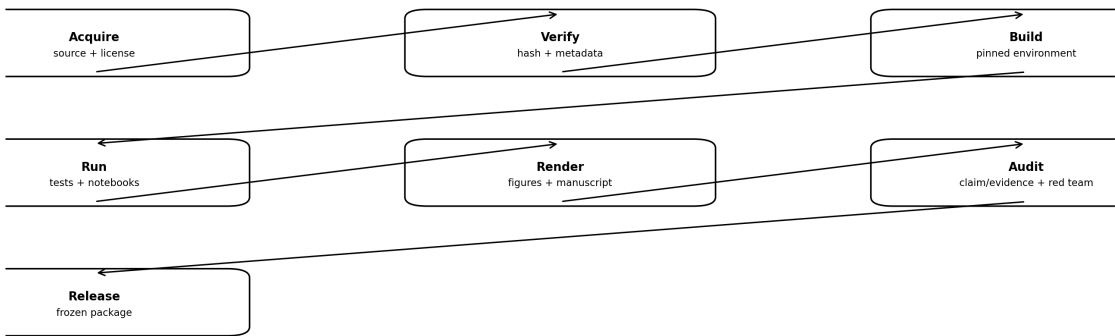


Figure 21.2: F23 repro.

What is measured? Detector-level observables and publicly documented response quantities.
What is inferred? Source or parent-particle properties that depend on model assumptions.
What would falsify it? A model-specific observation that is inconsistent with the predicted rate, spectrum, direction, companions, or causal structure.
Status of evidence. See the model cards and acceptance-gate register; absence of a calculation is labelled UNKNOWN rather than silently treated as a pass.

Research Roadmap and Conclusion

Evidence-status rule. Throughout this book, OBSERVED means directly tied to an experimental measurement; STANDARD means established physics with ordinary astrophysical assumptions; ACTIVE-HYPOTHESIS means a published mechanism under current investigation; PREPRINT-HYPOTHESIS means relevant literature exists but peer review is incomplete; LSC-SANDBOX means an original or highly speculative extension that has not earned promotion into the active physical model set.

The execution roadmap is intentionally staged. WP0 defines governance, source provenance, model identity, and freeze rules. WP1 reproduces the public KM3NeT event products. WP2 harmonizes detector exposures. WP3 audits multi-messenger counterparts. WP4 and WP5 build conventional hidden-source and cosmogenic kernels. WP6 handles PBHs. WP7 handles sterile/NSI/dark propagation. WP8 implements common inference. WP9 freezes prospective predictions.

The critical path is not “write more theory”. It is “make the competing models calculable under the same data model”. The largest blockers are public-response completeness, time- and direction-dependent effective areas for some transient scenarios, covariance and systematics across experiments, and the lack of a normalized quantum-geometry rate model.

The current scientific position is deliberately conservative. KM3-230213A is a secure cosmic UHE neutrino observation. Its parent energy is broadly uncertain and model-dependent. The source is unknown. Conventional hidden transients and cosmogenic production remain serious explanations. Recent IceCube point-source null searches tighten steady and recurrent source interpretations. Minimal four-dimensional PBH evaporation is under strong multi-messenger pressure. Modified PBHs, heavy dark matter, and detector-topography mechanisms are active but constrained. Black-bounce and black-to-white ideas are legitimate theoretical research topics but not yet event explanations.

The value of LSC-AW is therefore not that it supplies a new answer. It supplies a reproducible structure for refusing premature answers. That structure can become scientifically important if it is used to freeze predictions before the next UHE event and if it treats model failure as information rather than something to be edited away.

The final principle of this book is simple: where the data cannot select among models, say so. Where a parameter is unconstrained, say so. Where a source is unavailable, mark it unavailable. Where literature disagrees, reproduce both assumptions. And wherever possible, replace narrative plausibility with a quantity that can be calculated, reproduced, and eventually falsified.

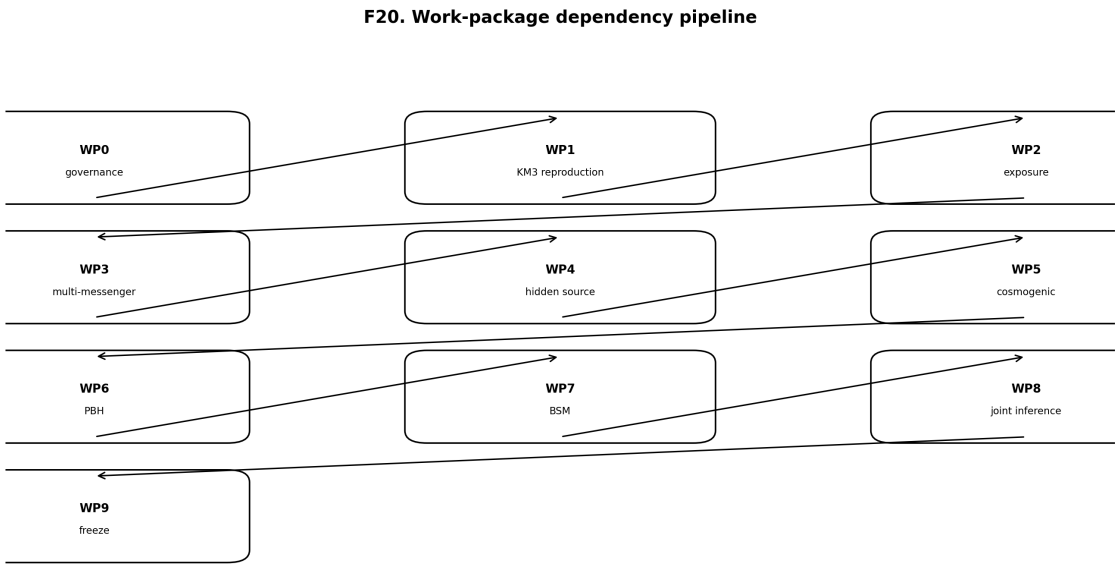


Figure 22.1: F20 wp pipeline.

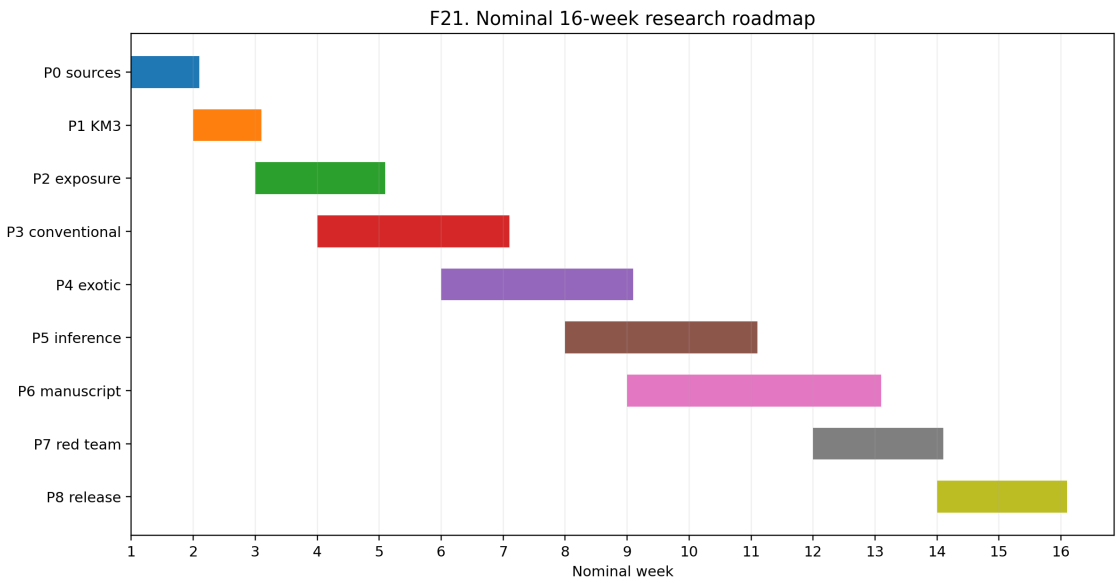


Figure 22.2: F21 gantt.

F24. Evidence ladder - status, not prestige

Higher is not "more exciting"; it is better supported.

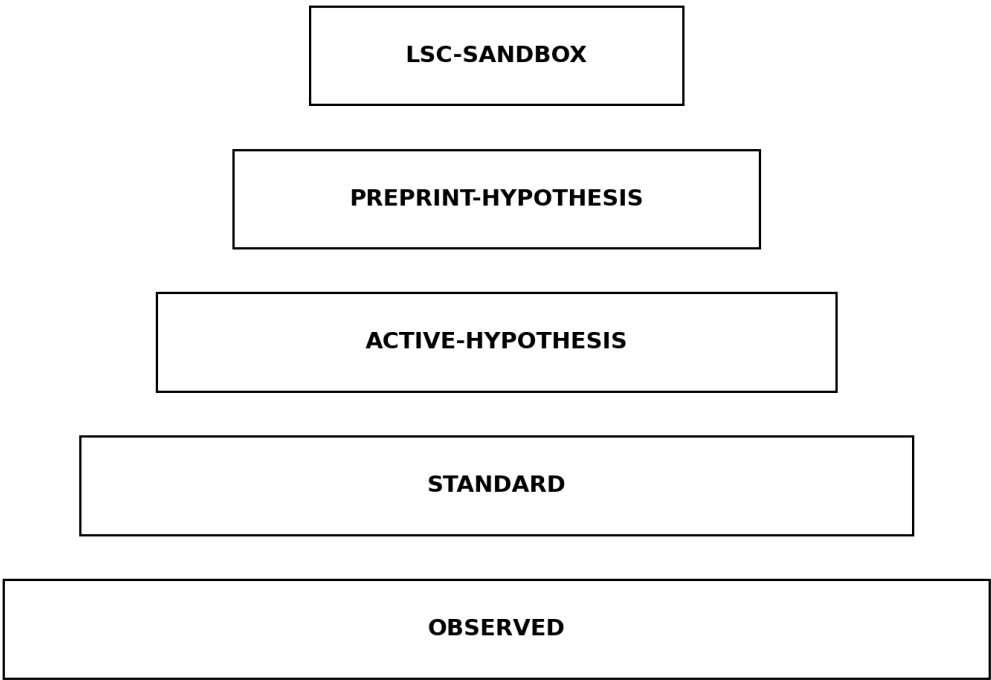


Figure 22.3: F24 evidence ladder.

What is measured? Detector-level observables and publicly documented response quantities.
What is inferred? Source or parent-particle properties that depend on model assumptions.
What would falsify it? A model-specific observation that is inconsistent with the predicted rate, spectrum, direction, companions, or causal structure.
Status of evidence. See the model cards and acceptance-gate register; absence of a calculation is labelled UNKNOWN rather than silently treated as a pass.

The Search for Counterparts

Evidence-status rule. Throughout this book, OBSERVED means directly tied to an experimental measurement; STANDARD means established physics with ordinary astrophysical assumptions; ACTIVE-HYPOTHESIS means a published mechanism under current investigation; PREPRINT-HYPOTHESIS means relevant literature exists but peer review is incomplete; LSC-SANDBOX means an original or highly speculative extension that has not earned promotion into the active physical model set.

The sky region around KM3-230213A was searched across gamma-ray, X-ray, optical, infrared, and radio catalogues. The original Nature paper identified twelve candidate extragalactic sources selected by several methods, but explicitly warned that the sky density of blazars makes chance associations common. The absence of a unique electromagnetic counterpart therefore cannot be turned into evidence for an exotic origin.

Subsequent work expanded candidate-source studies. Some blazars showed activity in particular bands, and population models demonstrated that hard-spectrum blazar populations can contribute a UHE neutrino flux without obviously exceeding the diffuse Fermi-LAT gamma-ray background. At the same time, direct point-source neutrino searches with KM3NeT, ANTARES, and IceCube did not find a significant excess at the event coordinates.

Localization remains important. The original error budget was dominated by systematic orientation uncertainty rather than the intrinsic statistical track fit. Improved calibration can shrink the search region and change the counterpart ranking. For LSC-AW, every counterpart analysis therefore stores the localization version used. Repeating a catalogue cross-match after a localization update is not double counting; treating both cross-matches as independent evidence would be.

A credible source association should satisfy more than angular coincidence. It should have plausible timing, enough hadronic power, a target photon or gas environment that can produce the neutrino, and an electromagnetic output consistent with observations. The strongest future association would combine all four: improved direction, contemporaneous source activity, a physically viable energy budget, and additional neutrino evidence.

What is measured? Detector-level observables and publicly documented response quantities.

What is inferred? Source or parent-particle properties that depend on model assumptions.

What would falsify it? A model-specific observation that is inconsistent with the predicted rate, spectrum, direction, companions, or causal structure.

Status of evidence. See the model cards and acceptance-gate register; absence of a calculation is labelled UNKNOWN rather than silently treated as a pass.

Rates, Rarity, and the One-Event Trap

Evidence-status rule. Throughout this book, OBSERVED means directly tied to an experimental measurement; STANDARD means established physics with ordinary astrophysical assumptions; ACTIVE-HYPOTHESIS means a published mechanism under current investigation; PREPRINT-HYPOTHESIS means relevant literature exists but peer review is incomplete; LSC-SANDBOX means an original or highly speculative extension that has not earned promotion into the active physical model set.

A model that can generate a 220 PeV particle is not automatically a model that can explain KM3-230213A. The missing quantity is rate. The universe may contain processes capable of enormous energies that are far too rare, too distant, or too weakly coupled to yield a detectable event.

For a Poisson process with expectation μ , the probability of observing one event is $\mu e^{-\mu}$. This simple formula illustrates why normalization matters. A model with $\mu \ll 1$ can still produce one event as a fluctuation, but a fit that sets $\mu = 1$ after observing one event has used the event to calibrate its rate and cannot then call the resulting agreement a prediction.

The official KM3NeT expected-events notebook demonstrates the correct calculation pattern: integrate flux times effective area over energy, multiply by livetime and sky/flavour factors appropriate to the published convention, and compare the expected count with the observation. Different source types require different geometrical factors; a short point-source burst cannot use a diffuse all-sky factor without justification.

This distinction is at the heart of the disagreement in the PBH literature. A population-based probability for at least one particle to intersect a large area around Earth is not the same object as the expected number of selected detector events after folding a time-dependent effective area. LSC-AW makes such normalization choices explicit in the model card.

Rate calculations are also the bridge from one event to future tests. A model that predicts one KM3NeT event per decade but also ten IceCube events is already in trouble. A model that predicts one event per century everywhere may require an unlikely fluctuation. A model that predicts a transient class at 0.1 per year can be tested by continued operation. Rarity is therefore not a rhetorical adjective; it is a quantitative property that must enter the likelihood.

What is measured? Detector-level observables and publicly documented response quantities.

What is inferred? Source or parent-particle properties that depend on model assumptions.

What would falsify it? A model-specific observation that is inconsistent with the predicted rate, spectrum, direction, companions, or causal structure.

Status of evidence. See the model cards and acceptance-gate register; absence of a calculation is labelled UNKNOWN rather than silently treated as a pass.

Model Cards

AW-M0 Background control. Mechanism: atmospheric or reconstruction background. Status: STANDARD control. Energy reservoir: cosmic-ray air showers or detector artefact. Main falsifier: consistency of geometry, track quality, and background simulation. Current assessment: strongly disfavoured for KM3-230213A but retained as a control.

AW-M1 Diffuse astrophysical tail. Status: STANDARD. Parameters: flux normalization, spectral index/breaks, flavour mix. Strength: minimal new physics. Weakness: low expected counts at UHE. Falsifier: repeated structured events inconsistent with smooth population statistics.

AW-M2 Generic transient. Status: STANDARD framework. Parameters: energy scale, width, sky position, duration, rate. Role: phenomenological bridge before source identification.

AW-M3 Hidden photohadronic source. Status: ACTIVE-HYPOTHESIS. Energy reservoir: accelerated hadrons. Key parameters: baryonic loading, proton index, target density, opacity, duration, luminosity function. Unique tests: source recurrence, cascade signatures, population rate.

AW-M4 Cosmogenic. Status: ACTIVE-HYPOTHESIS. Energy reservoir: UHECR propagation losses. Parameters: composition, proton fraction, source evolution, maximum rigidity, EBL/CMB. Unique tests: joint UHECR-neutrino-gamma fit.

AW-M5 Minimal 4D PBH. Status: ACTIVE-HYPOTHESIS / strongly constrained. Energy reservoir: PBH rest mass. Key tests: Hawking spectrum, companion photons and lower-energy neutrinos, burst rate. Current weakness: multi-messenger nulls.

AW-M6 Modified PBH. Status: ACTIVE/PREPRINT-HYPOTHESIS. Requires a separate card for every modification. Promotion requires explicit dynamics and a new falsifier.

AW-M7 Heavy relic/dark matter. Status: ACTIVE/PREPRINT-HYPOTHESIS. Parameters: mass, lifetime, channel, halo profile. Main constraints: gamma rays, other neutrino telescopes, anisotropy.

AW-M8 Sterile/NSI/dark propagation. Status: ACTIVE/PREPRINT-HYPOTHESIS. Parameters: mixing/coupling, mass splittings or mediator scales, matter dependence, decay length. Unique test: angular/topographic dependence across detectors.

AW-M9 Quantum geometry. Status: LSC-SANDBOX. Requirements before promotion: explicit geometry or effective Hamiltonian, normalized transition kernel, causal escape mechanism, source population, event rate, and independent falsifier.

Acceptance Gate Register

Every model is assessed against the following gates: Energy PDF, cross-experiment consistency, multi-messenger consistency, energy reservoir, conservation/backreaction, event-rate capability, trials/look-elsewhere accounting, prospective prediction, detector realism, and causal completeness where relevant. A PASS means the requirement is explicitly satisfied in a quantitative implementation. UNKNOWN is not a failure, but it prevents promotion to a strong explanatory claim. A model with a FAIL on a mandatory physical gate remains constrained or archived unless a new version is introduced.

The current qualitative status is summarized in Figure F18. This figure is not a numerical evidence score. Its purpose is to identify where the next calculation is needed. In particular, AW-M9 remains blocked on rate and causal implementation; AW-M5 is constrained by companions; AW-M3 and AW-M4 are mature enough for real rate calculations; and AW-M8 requires detector-specific population predictions.

Reproducibility Notes from Public KM3NeT Data

The public repository KM3NeT/KM3-230213A-data contains a bright-track all-flavour sky-averaged effective-area table spanning roughly 10^5 to 10^{11} GeV, selected event source data, and notebooks for expected-event calculations. The expected-events notebook uses a 335-day livetime and integrates the effective area against per-flavour flux models with the conventions documented in the repository.

The notebook output includes the following full-range expected-event counts for selected diffuse spectra: IceCube/NST 0.117431; IceCube/ESTES 0.052083; IceCube/HESE 0.019698; Aloisio cosmogenic model 0.469495; Ehlert cosmogenic model 0.242767; Muzio (2023) 0.431498; Fang source model 1.335689; Rodrigues (2020) source model 0.553614. These are reproductions of the published public notebook values, not new fits performed in this book.

The central 90 percent energy range used in that notebook is approximately 7.24×10^7 to 2.57×10^9 GeV. The small numerical differences from the rounded 72 PeV–2.6 EeV interval in the article reflect the machine values used in the notebook.

A future release-grade package should execute the notebooks from a clean environment rather than relying on value-level transcription. This manuscript therefore labels the table as SOURCE/REPRODUCED-FROM-PUBLIC-OUTPUT, not as an independent validation of the collaboration analysis.

LSC 6.5.0 Gallium Lineage Note

The historical LSC 6.5.0 object is included only to preserve programme identity. Its canonical profile is `EXACT_FINITE_A2/C/A` with one dimensionless parameter $\alpha_0 = -0.14388795521516184$ and an archived profile interval reported in the September 2026 research book. The model applies an exact finite dilation to the argument of a gallium capture cross-section table. The frozen policy is no-refit and `NOT_YET_VALIDATED`.

The historical book also preserves an adverse LSC 6.4.0 result and instructs that it must not be presented as a successful model. LSC 6.3.1 was abandoned before release. The version numbers belong to the gallium programme and are not UHE model versions.

The present LSC-AW programme borrows the freeze philosophy and evidence labelling but not the gallium parameter, detector tensor, or any low-energy numerical fit. Any future unification would require a new microscopic derivation connecting gallium-scale capture response and UHE production/propagation without post-hoc tuning.

Key Equations and Notation

Parent energy inference:

$$p(E_\nu \mid y, M) \propto p(y \mid E_\nu) \Phi_M(E_\nu).$$

Model-independent AW component:

$$\Delta\Phi_{\text{AW},\alpha} = A_\alpha f_E f_\Omega f_t.$$

Detector expectation:

$$\mu_{d,k} = \sum_{\alpha,\beta} \int \Phi_{\alpha,M} P_{\alpha \rightarrow \beta,M} A_{\text{eff},\beta}^{(d)} R_k^{(d)} dE d\Omega dt.$$

Point-process likelihood:

$$\mathcal{L} = e^{-\mu} \prod_i \int \lambda(E, \Omega, t) p(y_i \mid E, \Omega, t) dE d\Omega dt.$$

Hawking temperature:

$$T_H = \frac{\hbar c^3}{8\pi G M k_B}.$$

Quantum-geometry bookkeeping:

$$E_{\text{in}} + M_i c^2 = E_{\text{out}} + M_f c^2 + E_{\text{other}}.$$

The notation is intentionally modular. Source parameters are denoted θ_S , propagation parameters θ_P , detector parameters θ_D , and complete model parameters Θ_M .

Bibliographic Audit Priorities

Priority-zero sources are the Nature KM3NeT event paper, its correction, the KM3NeT Zenodo/Open Data package, and the public GitHub repository. Mandatory comparison families are IceCube EHE and point-source searches, Pierre Auger UHE-neutrino limits, KM3NeT source-population studies, cosmogenic UHECR analyses, event-specific PBH papers, BlackHawk v3.0, heavy dark-matter studies, sterile/NSI/dark-propagation papers, and black-to-white/black-bounce geometry foundations.

For every source the release package should record title, authors, year, peer-review status, DOI or arXiv identifier, version, URL, retrieval date, licence, code/data availability, chapters used, claims supported, limitations, and supersession status. Preprints and journal versions are deduplicated rather than counted as independent evidence.

Red-Team Checklist

The red team must attack: the E^{-2} energy prior; event-selection and effective-area assumptions; the difference between diffuse and transient exposure; counterpart trials factors; gamma/radio/IR/X-ray omissions; UHECR composition priors; PBH distance normalization; Hawking secondary spectra; field-of-view timing; dark-matter gamma constraints; sterile/NSI matter-column mapping; quantum-geometry causality; numerical integration stability; and any conclusion stronger than the supporting evidence.

A finding is not “resolved” because the authors disagree with it. It is resolved only by a primary source, a reproduced calculation, a corrected model, or a documented downgrade in the claim. This rule is essential in a research programme where many hypotheses were generated after the event became known.

Prospective Freeze Template

Each frozen model release should contain: model ID and version; parameter values and priors; source dataset list; code commit; environment hash; predicted next-event energy distribution; predicted directional distribution; predicted waiting-time distribution; expected detector ratios; expected electromagnetic companions; explicit rejection thresholds; and a statement of which future dataset will be used for scoring.

The package is hashed and timestamped. When the target data become available, the score is computed before any refit. If the model is updated afterward, the new version receives a new identity and must earn validation on different data.

Archived Mechanisms

The following ideas are retained only for provenance unless a future derivation changes their status: classical escape from a permanent event horizon; tachyonic escape; arbitrary extraction of hundreds of PeV from “quantum foam” without a defined operator and backreaction; literal information cloning as particle duplication; and traversable wormholes without an explicit metric, stress-energy source, stability analysis, and transition probability.

Archival does not mean metaphysical impossibility. It means that the idea, in its current form, does not satisfy the minimum requirements of a quantitative event model.

Selected Data Tables

KM3-230213A observational summary

Quantity	Reported value / status
Event time	13 Feb 2023, 01:16:47 UTC
ARCA configuration	21 detection lines operating
Triggered hits	28,086
Triggered PMTs	3,672
Muon energy	120^{+110}_{-60} PeV; 90% interval about 35–380 PeV
Parent neutrino energy under E^{-2}	median 220 PeV; 68% 110–790 PeV; 90% 72 PeV–2.6 EeV
Equatorial direction	RA 94.3°, Dec -7.8° (J2000)
Containment	68% radius 1.5° ; 99% radius 3.0°
Standalone flux interpretation	Tension with IceCube/Auger; upward-fluctuation interpretation about 2.2σ in KM3NeT analysis

Public expected-event examples from the KM3NeT notebook

Flux/model	Total events	Events in central 90% energy range
IceCube/NST	0.117431	0.014254
IceCube/ESTES	0.052083	0.003429
IceCube/HESE	0.019698	0.000509
Aloisio cosmogenic	0.469495	0.015867
Ehlert cosmogenic	0.242767	0.211117
Muzio (2023) cosmogenic	0.431498	0.241579
Fang source model	1.335689	0.471527
Rodrigues (2020) source model	0.553614	0.260915
Winter TDE	0.444677	0.054590

These values are taken from the output embedded in the official public KM3NeT expected-events notebook. They are included as reproduction anchors, not as new model fits.

Literature Status Matrix

Primary event and detector sources

Source family	Status	Role in LSC-AW
KM3NeT Nature event paper	Peer-reviewed	Primary detector observation, event topology, energy reconstruction, sky localization, standalone flux.
Nature correction	Peer-reviewed correction	Corrects author metadata; scientific conclusions unchanged.
KM3NeT Open Data / Zenodo / GitHub	Official public data	Reproduction anchor for skymap, hit distributions, effective area, flux-model comparisons, and expected-event tables.
IceCube EHE search	Peer-reviewed	External UHE neutrino constraint and cross-experiment baseline.
IceCube point-source search toward KM3-230213A	2026 preprint	Direct directional test of persistent/flaring neutrino emission; no significant evidence reported.
Pierre Auger UHE-neutrino searches	Peer-reviewed / official releases	Independent exposure and upper limits at ultrahigh energies.

Astrophysical and cosmogenic models

Model family	Evidence status	Central scientific issue
Diffuse astrophysical tail	STANDARD	Whether one event can be a rare draw from a continuation of the known flux.
Hidden transient / blazar / AGN	ACTIVE-HYPOTHESIS	Can a short-lived, gamma-opaque hadronic source produce the event without violating IceCube and multiwavelength constraints?
Long-GRB blast waves	ACTIVE-HYPOTHESIS	Baryon loading and circum-burst density constrained through joint detector exposure.
Cosmogenic UHECR	ACTIVE-HYPOTHESIS	Requires a hard UHE proton component/source evolution while satisfying composition and gamma-cascade constraints.
Galactic conventional source	CONSTRAINED	No compelling known Galactic counterpart in the localization region; energy budget is demanding.

Exotic models

Model family	Evidence status	Central scientific issue
Minimal 4D PBH evaporation	ACTIVE-HYPOTHESIS / STRONGLY CONSTRAINED	Companion gamma and lower-energy neutrino signals are difficult to hide.
Modified PBH	ACTIVE/PREPRINT	Must state the specific modification and calculate its new spectrum, rate, and constraints.
Heavy dark matter	ACTIVE/PREPRINT	Detector-folded fits are possible but preferred regions are in tension with other messenger limits.
Sterile/NSI/dark propagation	ACTIVE/PREPRINT	Predicts detector/matter-column dependence; must still explain the UHE incident flux or parent state.
Black-bounce / black-to-white	LSC-SANDBOX / theory foundation	Geometry exists in the literature, but event-level normalized production and rate remain unestablished.

Discriminating-Observable Matrix

Observable	Hidden transient	Cosmogenic	PBH	Heavy DM	Topography/BSM
Repeat from same direction	possible	not required	very unlikely per burst	not point-like in halo baseline	source-model dependent
Short time clustering	expected for flare	weak	strong terminal burst	weak	conversion/decay dependent
Gamma companion	may be absorbed	cascades diffuse	strong in minimal model	channel dependent	usually secondary
UHECR relation	source hadrons	direct parent population	none required	none required	source dependent
Galactic anisotropy	source dependent	weak	halo population possible	expected halo contribution	geometry dependent
Detector-site dependence	ordinary exposure	ordinary exposure	ordinary exposure	ordinary exposure	potentially defining signal
Unique rate prediction	source population	UHECR population	PBH mass function/rate	lifetime + halo density	requires flux + conversion model

The table is intentionally not scored. A model can be highly predictive in one observable and weak in another. The purpose is to identify which next measurement is discriminating rather than to produce an arbitrary total score.

Figure Provenance Register

ID	Figure	Provenance status
F01	Detector schematic	Original conceptual drawing; not to scale.
F02	Energy inference	Constructed from published muon and parent-neutrino intervals; no digitized likelihood used.
F03	Energy landscape	Selected published reference scales; explanatory only.
F04	Sky localization	Candidate coordinates from the official KM3NeT public Figure 4 source CSV; localization radii from the Nature paper.
F05–F08	Architecture diagrams	Original LSC-AW conceptual diagrams.
F09	KM3NeT effective area	Directly plotted from official public JSON in KM3NeT/KM3-230213A-data.
F10–F11	Source mechanisms	Original conceptual schematics.
F12	PBH mass-temperature relation	Calculated from the Schwarzschild Hawking scaling for explanatory purposes.
F13	PBH distance dilution	Generic D^{-2} fluence curve with representative distances from two published PBH normalization arguments.
F14	Geometry comparison	Schematic only; explicitly not BlackHawk numerical output.
F15–F16	Causality/topography	Original conceptual schematics.
F17–F18	Evidence/gate maps	Qualitative research-status diagrams, not probabilities or statistical scores.
F19–F24	Prediction/workflow/evidence diagrams	Original LSC-AW process graphics.

Quantitative geometry in F02, F04, F09, F12 and F13 is generated from stated values or equations. The remaining figures are explicitly conceptual and must not be cited as measured distributions.

Scientific Test Status Register

This manuscript defines the test suite required by the execution specification, but it does not mislabel every defined test as completed. The status below separates source-level reproduction from future implementation work.

ID	Test	Status in this book	Comment
T00	Provenance completeness	PARTIAL	Primary sources and major public data are identified; full release manifest still required.
T01	KM3 event reproduction	SOURCE-ANCHORED	Public notebook outputs and effective-area files inspected; full clean-room execution not claimed here.
T02	Energy-posterior reweighting	NOT RUN	Requires model-conditioned detector response or released posterior machinery.
T03	Coordinate/time consistency	SOURCE-ANCHORED	Published event time and coordinates used consistently.
T04	Atmospheric/background sanity	SOURCE-ANCHORED	Published rates and range arguments summarized, not independently simulated.
T05	Exposure harmonization	PARTIAL	KM3 public area used; complete joint direction/time-dependent harmonization remains a work package.
T06	Hidden-transient energy budget	LITERATURE-BASED	Published blazar/GRB work summarized; no new population fit claimed.
T07	Cosmogenic propagation	LITERATURE-BASED	Public model event counts and 2026 literature summarized.
T08	BlackHawk reproduction	NOT RUN	BlackHawk v3 capability verified from source paper; local benchmark run still required.
T09	PBH companions	LITERATURE-BASED	Supportive and critical published calculations compared.
T10	Sterile/NSI topography	LITERATURE-BASED	Mechanism reviewed; independent detector-column simulation not run.

T11	Dark-matter messengers	LITERATURE-BASED	2026 detector-folded analysis summarized.
T12	QG conservation	CONCEPTUAL PASS / RATE UNKNOWN	Energy bookkeeping stated; normalized kernel not available.
T13	Likelihood normalization	FORMALISM ONLY	Equations specified; common event-level implementation remains future work.
T14–T16	Prior sensitivity / trials / PPC	NOT RUN GLOBALLY	Required before quantitative model preference.
T17	Next-event freeze	TEMPLATE ONLY	No physical freeze issued for incomplete models.
T18	Independent rebuild	PARTIAL	PDF and figures rebuilt locally; external clean-room scientific rebuild not claimed.
T19	Claim-evidence audit	MANUSCRIPT-LEVEL	Claims are deliberately downgraded where computations are absent.
T20	Figure integrity	PASS FOR THIS PDF	All included figures generated from local code; conceptual figures labelled.

This register is central to the integrity of the book. “Not run” is a scientific status, not an embarrassment. A future execution package should replace these entries only after fresh reproducible evidence exists.

Editorial and Reviewer Guide

A reviewer can evaluate this manuscript in three layers. First, the observational layer asks whether the detector facts, energy intervals, coordinates, and source-search outcomes are stated correctly. Second, the inference layer asks whether each model is compared under compatible exposure and multi-messenger assumptions. Third, the programme layer asks whether LSC-AW adds scientific value as a reproducible architecture rather than merely renaming a literature review.

Reviewers should pay particular attention to category errors. A detector-level track energy must not be treated as an exactly measured source-neutrino energy. A population probability must not be substituted for a selected detector-event likelihood. A mathematical regular black-hole metric must not be presented as evidence that the event came from a quantum bounce. A model's ability to avoid an external constraint because it makes no prediction there must not be counted as positive support.

The most valuable referee criticism would identify a missing primary dataset, an inconsistent effective-area convention, a model-specific companion signal that has been omitted, or a claim whose status is too strong. Such criticism should result in either a corrected calculation or an explicit downgrade, not rhetorical defence.

Suitable publication routes differ by format. The complete volume is closer to a research monograph or long technical review than a standard short journal article. A derivative methods paper could focus on the common source-propagation-detector likelihood and provenance architecture. A derivative phenomenology paper could focus on one reproduced model family, such as PBH companion constraints or cosmogenic population inference. The full book should not be forced into an article format that requires removing the uncertainty and provenance material that gives it scientific value.

Limitations and Open Problems

The most important limitation is the single-event nature of the trigger. One event can constrain models with strong companion predictions, but it cannot by itself establish a new population. The second limitation is public detector-response completeness. KM3NeT provides valuable public data, yet some transient comparisons require direction- and time-dependent response information that is not fully represented by a sky-averaged effective area.

The third limitation is model heterogeneity. A cosmogenic model is a population model, a PBH burst is a transient compact-object model, heavy dark matter is a decay model, and LSC-AW Core is an inference architecture. Numerical ranking among such objects is meaningful only after each has a fully specified likelihood, priors, nuisance structure, and predictive domain.

The fourth limitation is that the quantum-geometry branch is not yet an event-rate model. The existence of regular metrics and black-to-white constructions is important theoretical context, but the LSC-specific transition kernel remains undefined. This book therefore refuses to promote that branch beyond sandbox status.

The fifth limitation is that the historical gallium LSC programme remains separate. Its 6.5.0 response deformation is reproducible but not validated, and no microscopic derivation currently links its fitted scale to UHE neutrino production. A future unification would have to explain both sectors with new independent predictions; visual similarity between equations is not enough.

Open problems include: reconstructing model-conditioned parent-energy posteriors; harmonizing KM3NeT/IceCube/Auger transient exposures; reproducing BlackHawk event-specific spectra in a common environment; building a fully joint likelihood with trials corrections; obtaining improved localization; and freezing model-specific next-event predictions before additional UHE events are inspected.

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Final Scientific Position

LSC does not assume the answer to KM3-230213A. It establishes a common, reproducible arena in which conventional astrophysical sources, cosmogenic propagation, primordial-black-hole emission, dark-sector physics, sterile-neutrino mechanisms, and quantum-geometry hypotheses must explain the same observation while surviving the same null results and multi-messenger constraints. Where the data cannot select among models, this book says so. Where a model cannot yet calculate a rate, it remains a sandbox proposal.